Chapter 5

MULTISTAGE, POWER, AND DIFFERENTIAL AMPLIFIERS

CHAPTER OUTLINE

5-1 Coupled Amplifiers

5-2 Power Amplifiers

5-3 Differential Amplifiers

5-4 Frequency Response of Amplifiers

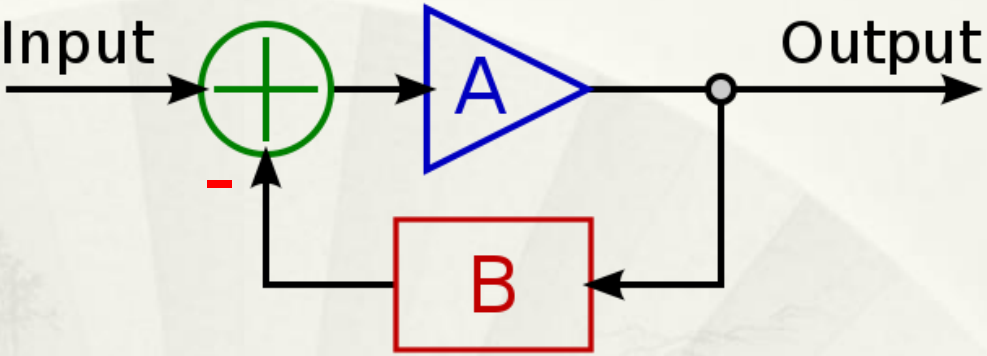
5-5 Negative Feedback in Amplifiers

SCIENCE HIGHLIGHT

- **Negative Feedback**



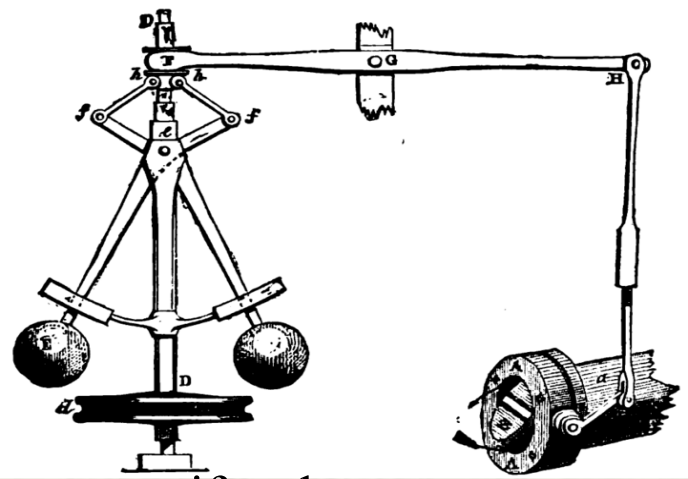
Harold Stephen Black
1898.4 – 1983.12



Negative feedback occurs when the output of a system acts to oppose changes to the input of the system, with the result that the changes are attenuated.

SCIENCE HIGHLIGHT

- **Negative Feedback**



centrifugal governor



sweating



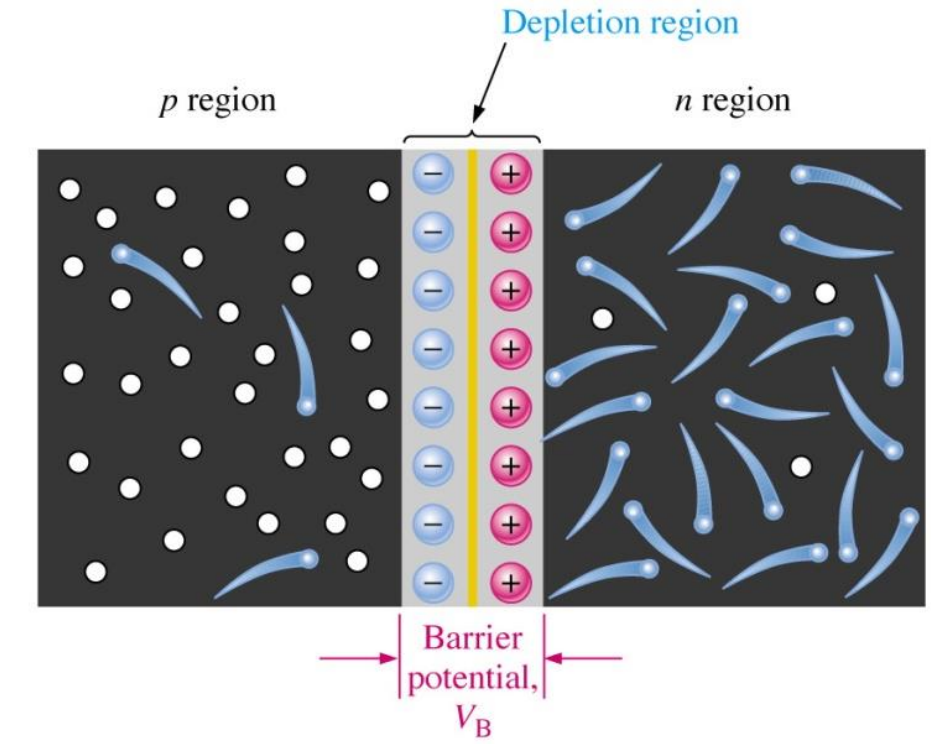
air-condition

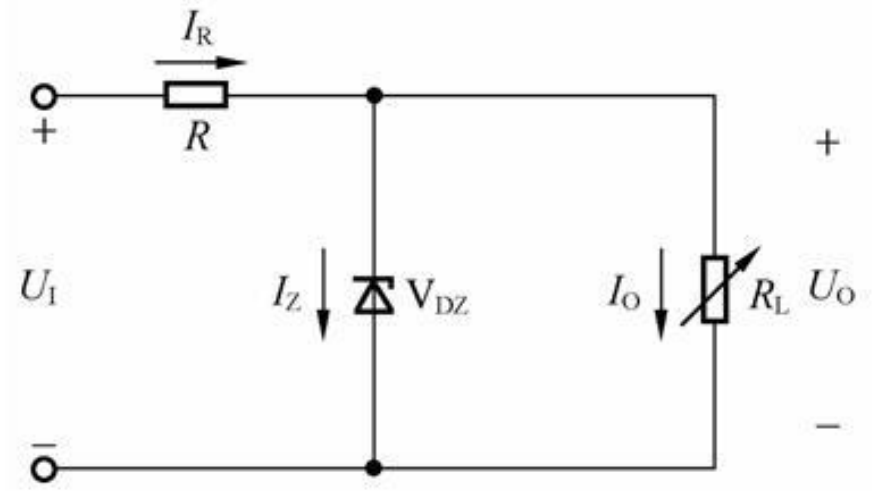


parachute

SCIENCE HIGHLIGHT

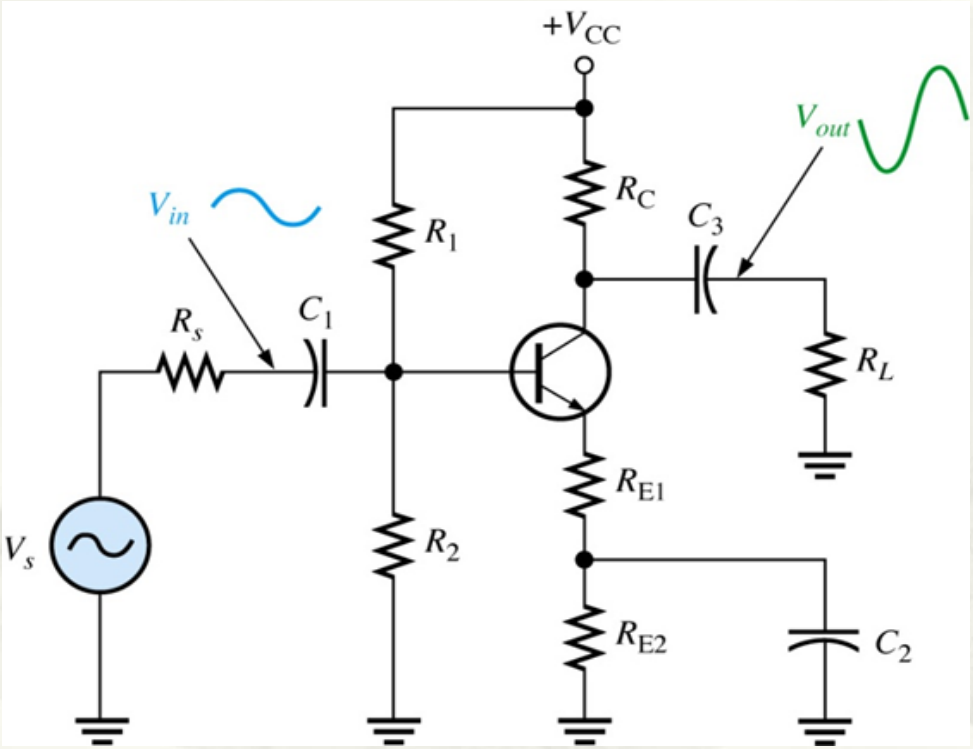
- **Negative Feedback**





SCIENCE HIGHLIGHT

- **Negative Feedback**



If one, or both of these opposing influences are non-linear, equilibrium (stable) point(s) result.

5-5 Negative Feedback in Amplifiers

5-5-1 Basic Concepts

5-5-2 Category

5-5-3 Effects on Performance

5-5-4 Analysis Methods Based on Block Diagram

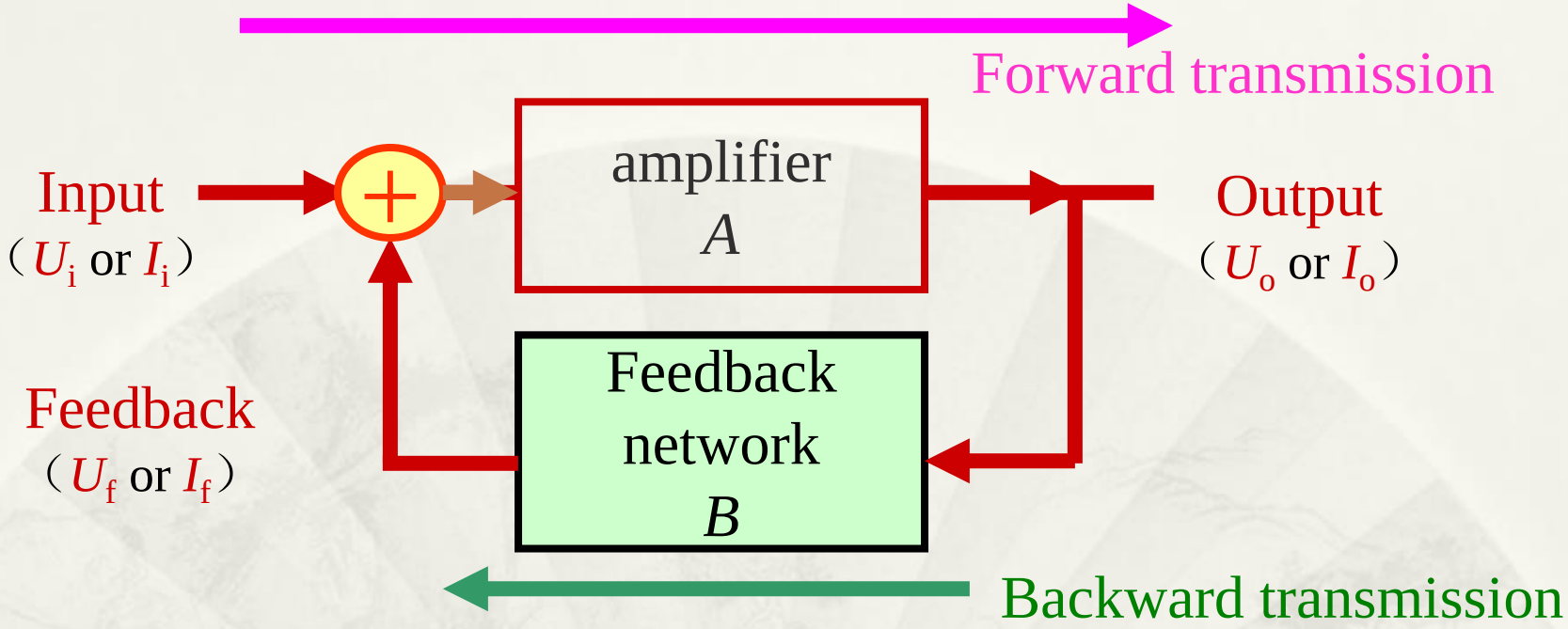
5-5-5 Closed-loop Stability of Multistage Amplifiers

5-5-6 Brief Summary

5-5-1 Basic Concepts

- Definition of Feedback
- Category of Feedback
- Idea Model for Single-Loop Feedback
- Basic Feedback Equations

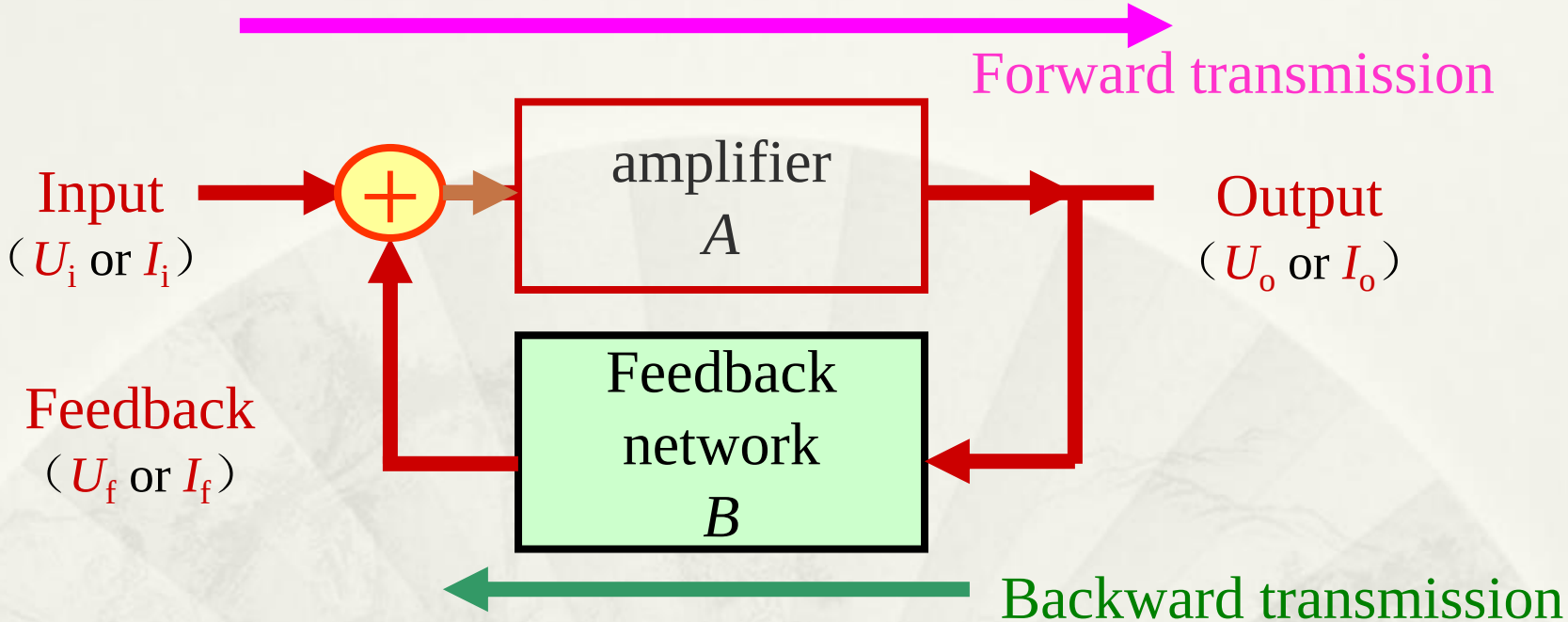
Definition of Feedback



feedback:

- Part or all of the outputs are taken out and sent back to the input loop of the amplifier.
- After adding or subtracting the original input, they are applied to the input end of the amplifier.

Definition of Feedback

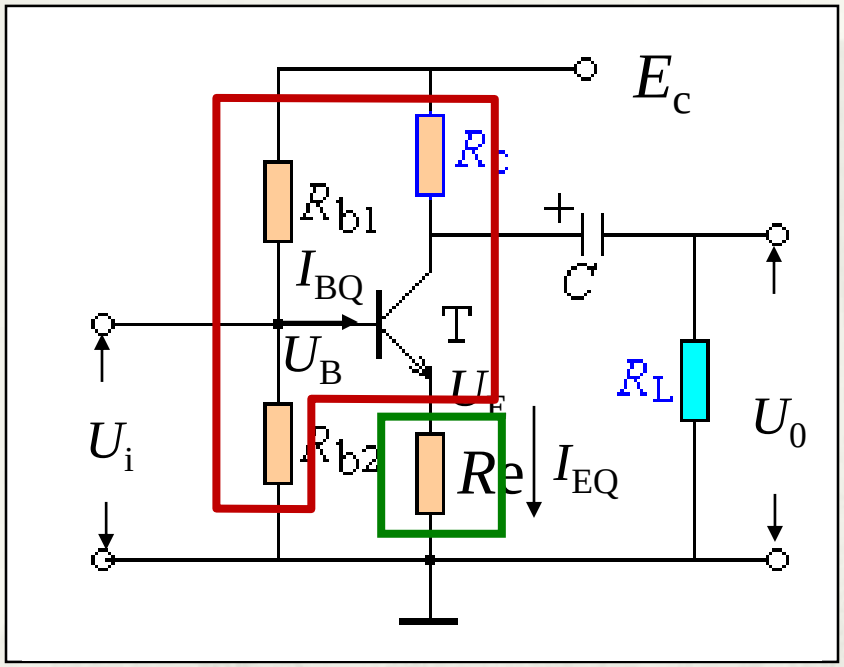


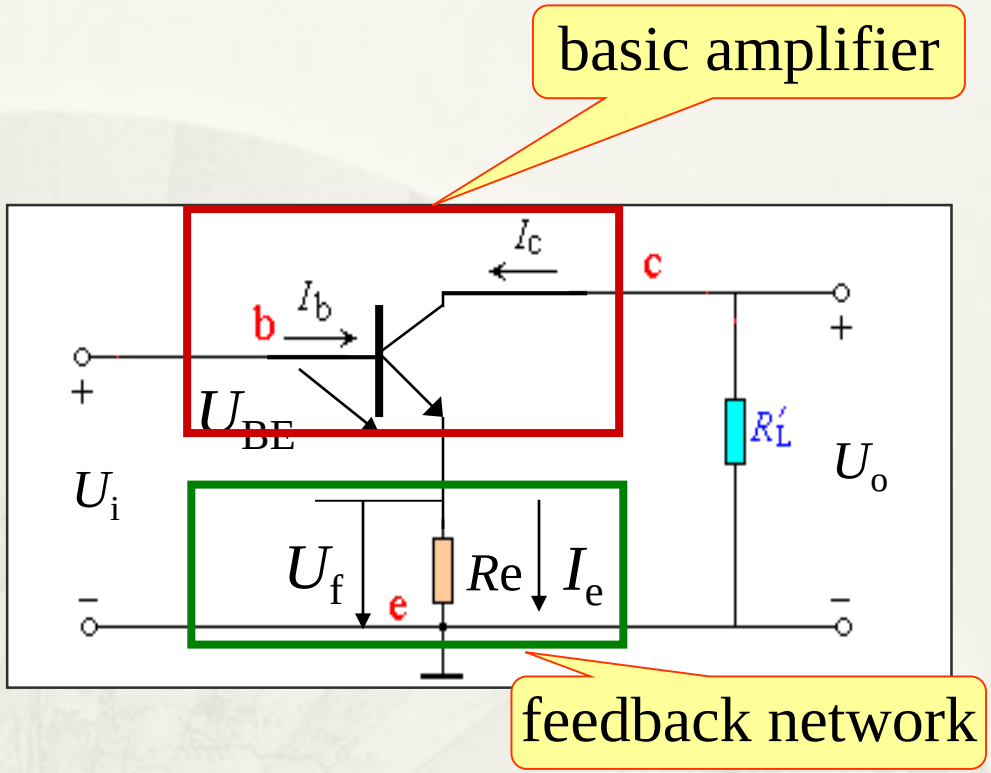
open-loop: an amplifier without feedback

closed-loop: an amplifier with feedback

Definition of Feedback

- **Example**





Q-point is stable:

$$I_{CQ}\downarrow \leftarrow I_{BQ}\downarrow \leftarrow U_{BE}=U_B\text{-}U_E\downarrow$$

$$T\uparrow \rightarrow U_{BE}\downarrow, \beta\uparrow \rightarrow I_{CQ}\uparrow \rightarrow I_{EQ}\uparrow \rightarrow U_E=I_{EQ}R_e\uparrow$$

Category of Feedback

Different Categories for Different Starting Points

Polarity of feedback

- positive: feedback signal increases AMP's net input signal X_{di}
- negative: feedback signal reduces AMP's net input signal X_{di}

Property of feedback

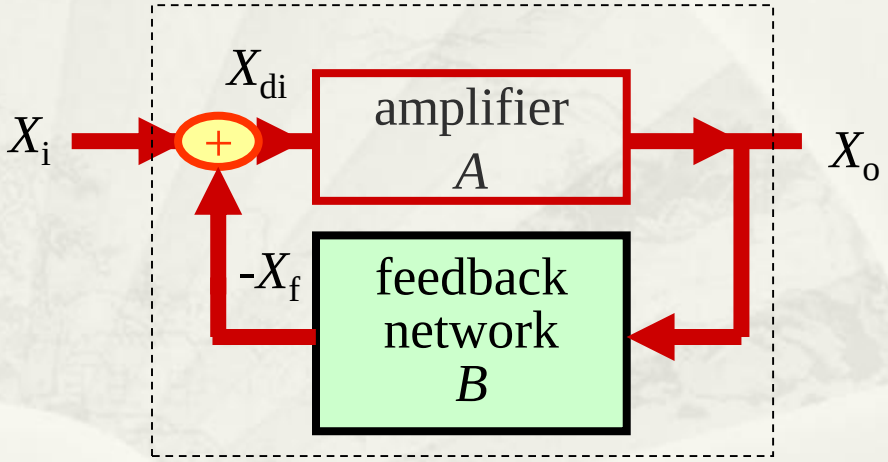
- DC
- AC
- mixed

Sampling signal of feedback

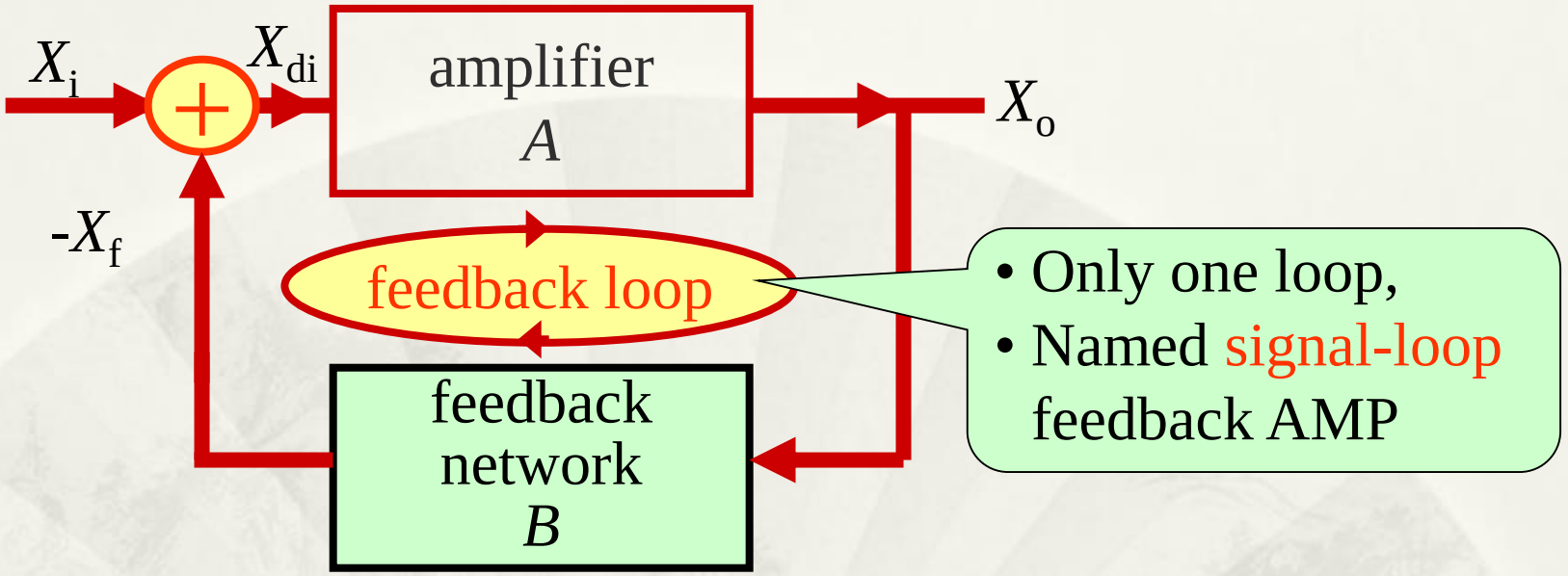
- voltage
- current

Introducing mode of feedback in input

- series
- parallel



Idea Model for Single-Loop Feedback

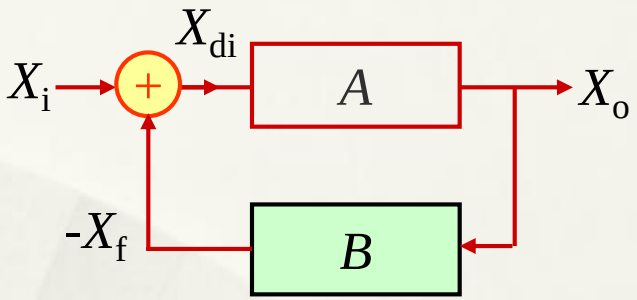


- **Idea model**

  1. Signals in forward transmission only pass through *A*
  2. Signals in backward transmission only pass through *B*

Basic Feedback Equations

- **For negative feedback**



open-loop gain $A = \dfrac{X_o}{X_{di}}$

for negative feedback AMPs: $X_{di} = X_i - X_f$

Feedback coefficient $B = \dfrac{X_f}{X_o}$

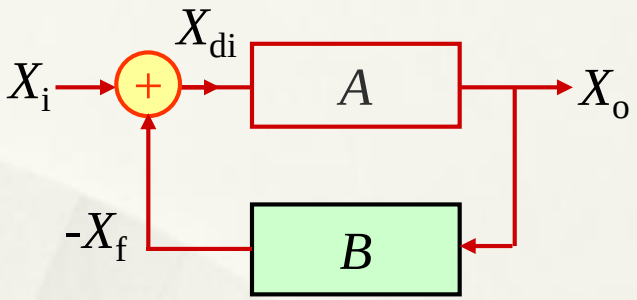
closed-loop gain $A_f = \dfrac{X_o}{X_i} \rightarrow A_f = \dfrac{A}{1 + AB}$

loop gain AB

feedback depth $F = 1 + AB$

Basic Feedback Equations

For negative feedback



In **deep** negative feedback

$F = 1 + AB >> 1$ or $AB >> 1$

$$A_f = \frac{A}{1+AB} \approx \frac{A}{AB} = \frac{1}{B}$$

- The closed-loop gain is approximately equal to the reciprocal of the feedback coefficient
- basically independent of the parameters of the active devices.
- Generally, the feedback network is composed of passive components, and its stability is better than that of active components.
- Therefore, the amplification factor of deep negative feedback is relatively stable.

5-5 Negative Feedback in Amplifiers

5-5-1 Basic Concepts

5-5-2 Category

5-5-3 Effects on Performance

5-5-4 Analysis Methods Based on Block Diagram

5-5-5 Closed-loop Stability of Multistage Amplifiers

5-5-6 Brief Summary

5-5-2 Category of Negative Feedback

- Current Series Negative Feedback
- Voltage Series Negative Feedback
- Current Parallel Negative Feedback
- Voltage Parallel Negative Feedback

Current Series Negative Feedback

Determine the type of feedback

step 1: find feedback network

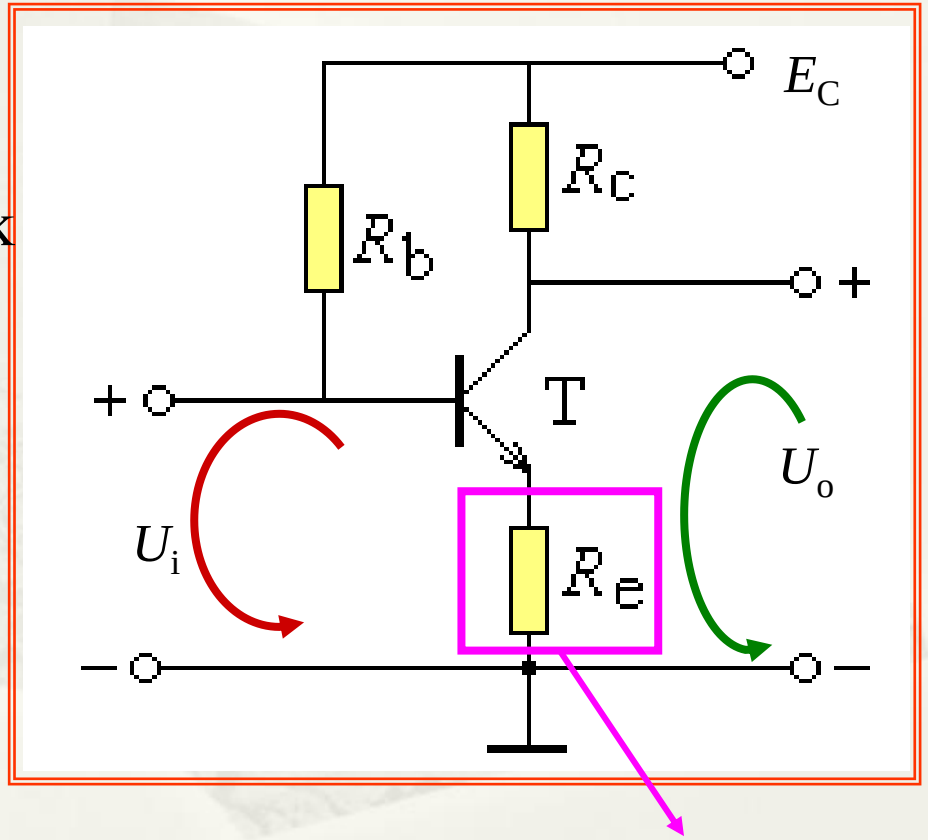
-- related to input loop and output loop

step 2: determine the type of feedback

① voltage or current **?**

voltage feedback: proportional to output voltage

current feedback: proportional to output current



Here, $U_f = I_o R_e$

↳ it's current feedback

Current Series Negative Feedback

Determine the type of feedback

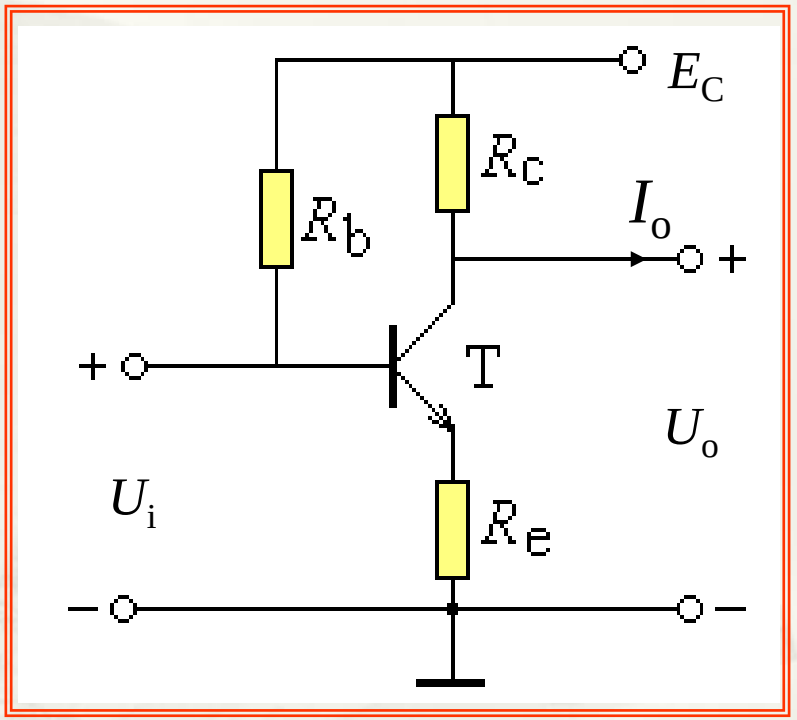
step 1: find feedback network

step 2: determine the type of feedback

① voltage or current **?**

Simple method

When the output voltage is zero, it's voltage feedback if the feedback signal is zero. Otherwise, it's current feedback.



Current Series Negative Feedback

Determine the type of feedback

step 1: find feedback network

step 2: determine the type of feedback

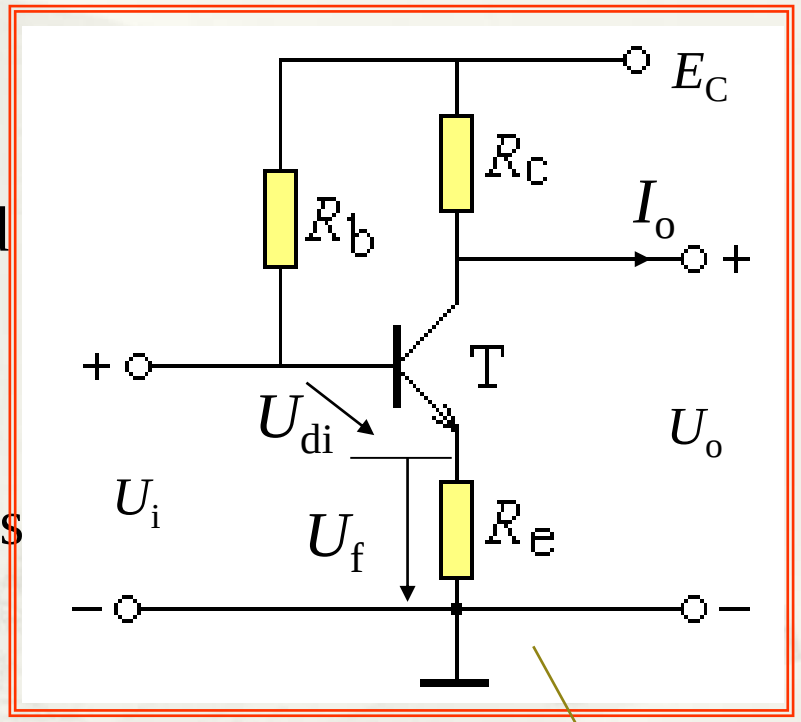
② series or parallel **?**

It's parallel feedback if the feedback and the input signals are added to the same terminal of the AMP input loop.

-- the relationship between the feedback signal and the input signals is the addition or subtraction of current.

Otherwise, it's series feedback.

-- the relationship between the feedback signal and the input signal is the addition or subtraction of voltage.



series feedback

Current Series Negative Feedback

Determine the type of feedback

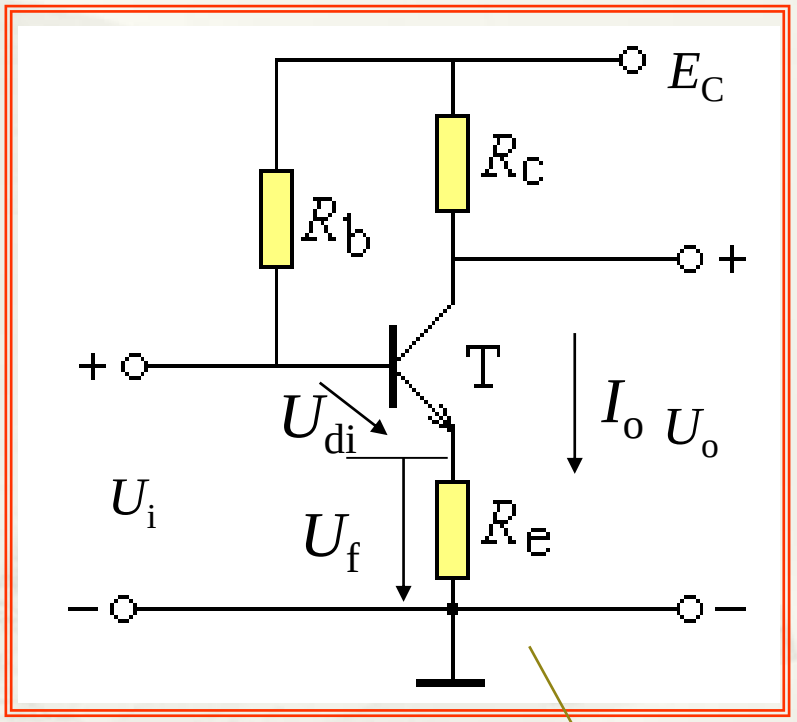
step 1: find feedback network

step 2: determine the type of feedback

② series or parallel **?**

For BJT

- If the feedback and the input signals are added to the base or emitter of the transistor together , it's parallel feedback.
- If one is added to the base and another is added to the emitter, it's series feedback.



series feedback

Current Series Negative Feedback

Determine the type of feedback

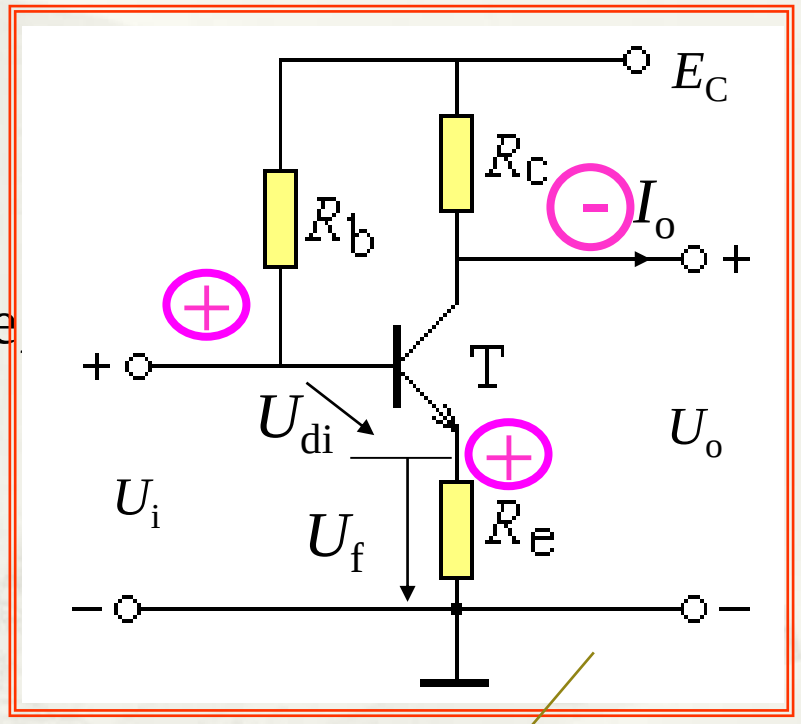
step 1: find feedback network

step 2: determine the type of feedback

③ positive or negative ?

Instantaneous polarity method:

- Assume the polarity of the input voltage
- Determine the instantaneous polarity of relevant nodes
- Finally, determine the instantaneous polarity of the feedback signal.
  -- it's negative feedback if the polarity of the feedback signal let the net input reduced. Otherwise, it's positive polarity.



$$U_{di} = U_i - U_f \quad < U_i$$

it's negative feedback

Current Series Negative Feedback

Determine the type of feedback

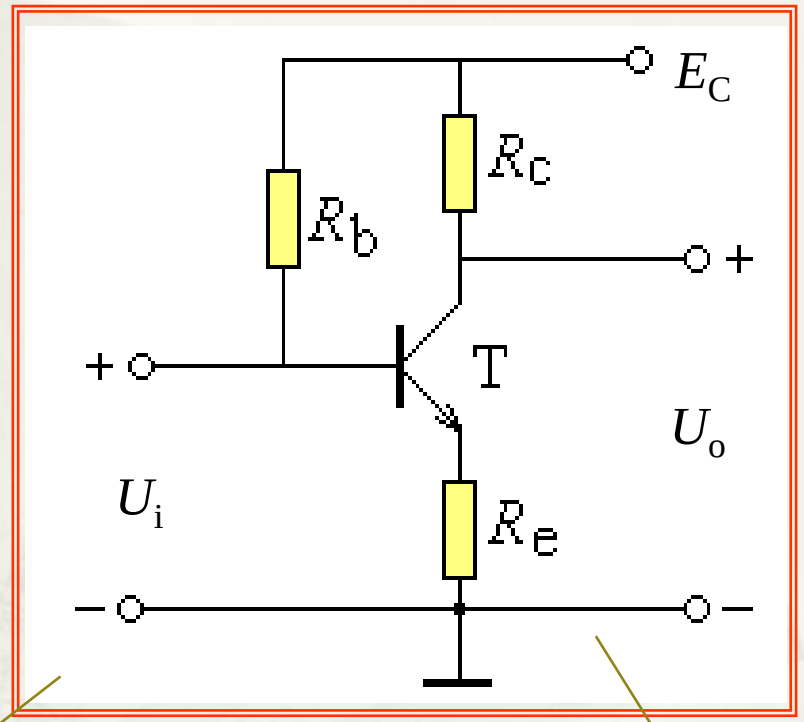
step 1: find feedback network

step 2: determine the type of feedback

④ DC or AC **?**

Capacitor observation method:

- It's AC feedback if there is coupling capacitors in the feedback path;
- It's DC feedback if there is bypass capacitors in the feedback path;
- It's DC-AC mixed feedback if there is no capacitors in the feedback path.

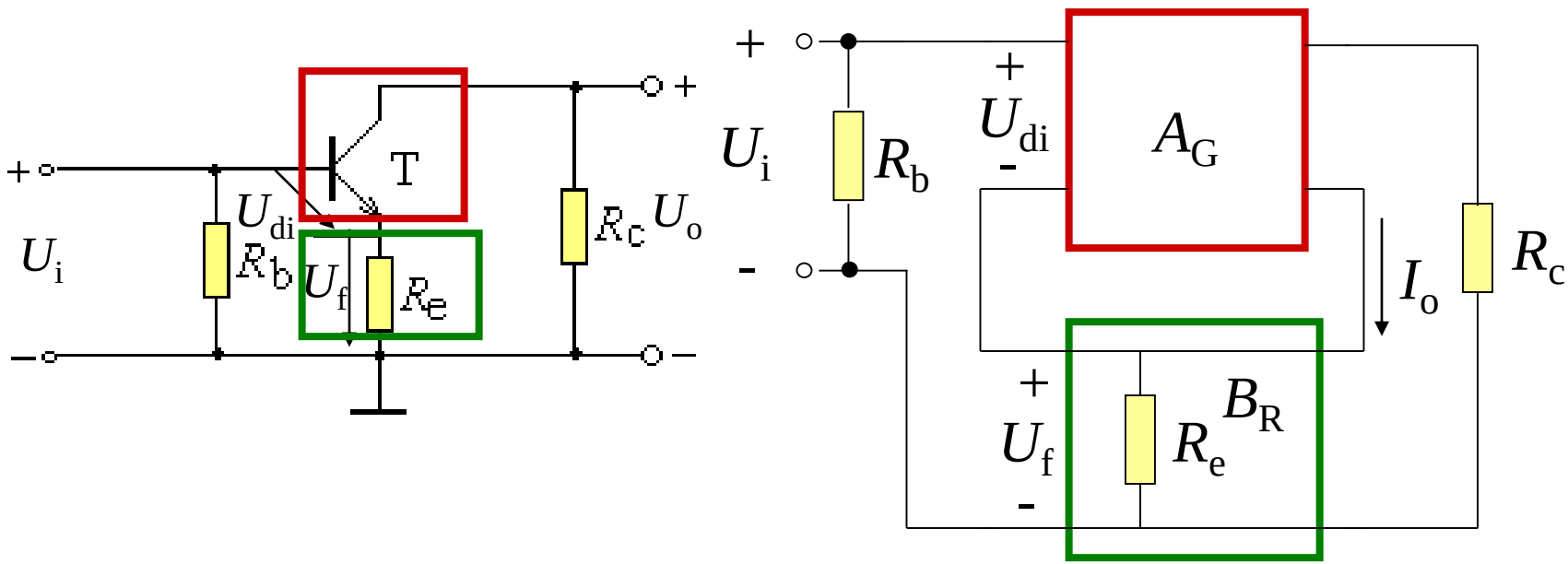


it's DC-AC mixed feedback

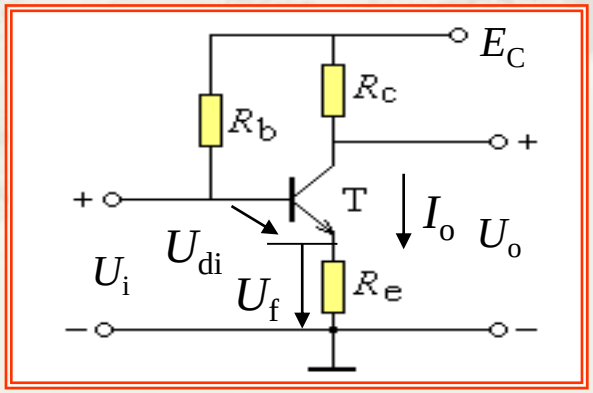
Overall, the feedback introduced by the feedback component R_e is a DC-AC mixed current series negative feedback

Current Series Negative Feedback

- **Block Diagram** -- based on AC path



In current series negative feedback, feedback comes from output current and adjusts input voltage, so it is better to describe the gain of the feedback configuration with output current and input voltage.



Current Series Negative Feedback

Gain and Feedback Coefficient

open-loop gain

$$A_G = \frac{I_o}{U_{di}}$$

-- conductance dimension
-- transconductance gain

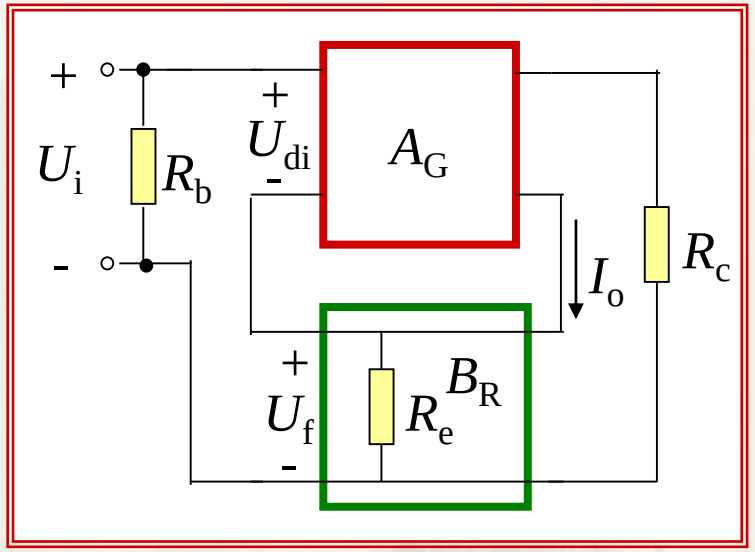
closed-loop gain

$$A_{Gf} = \frac{I_o}{U_i}$$

feedback coefficient

$$B_R = \frac{U_f}{I_o}$$

-- resistance dimension
-- transresistance coefficient



closed-loop voltage gain

$$A_{Uf} = \frac{U_o}{U_i} = A_{Gf} R_c$$

feedback equation

$$A_{Gf} = \frac{A_G}{1 + A_G B_R}$$

feedback depth

$$F = 1 + A_G B_R$$ -- dimensionless

Voltage Series Negative Feedback

Determine the type of feedback

step 1: find feedback network

—R_{e1}, R_{e2}, R_f /R_{e1}

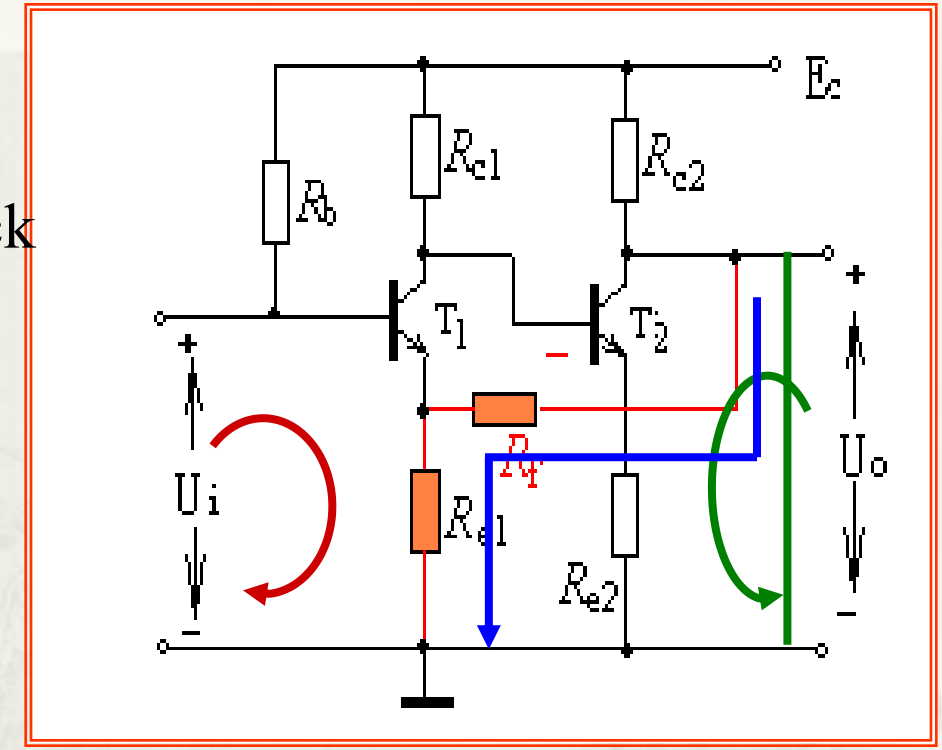
step 2: determine the type of feedback

① voltage or current ?

- short the output voltage to ground
- feedback disappears. it's voltage feedback.

② series or parallel ?

- input and feedback signals are added to two terminals of transistor. it's series feedback.



Voltage Series Negative Feedback

Determine the type of feedback

step 1: find feedback network

—R_{e1}, R_{e2}, R_f /R_{e1}

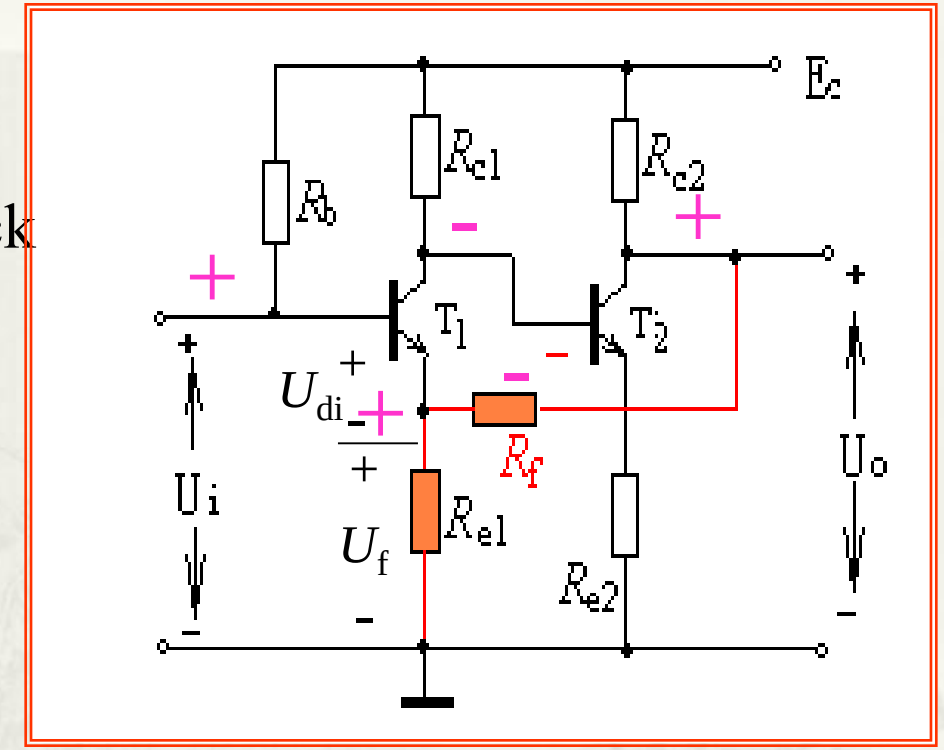
step 2: determine the type of feedback

③ positive or negative **?**

- Assume the polarity of input voltage is positive
- The polarity of feedback voltage is positive, it's negative feedback

④ DC or AC **?**

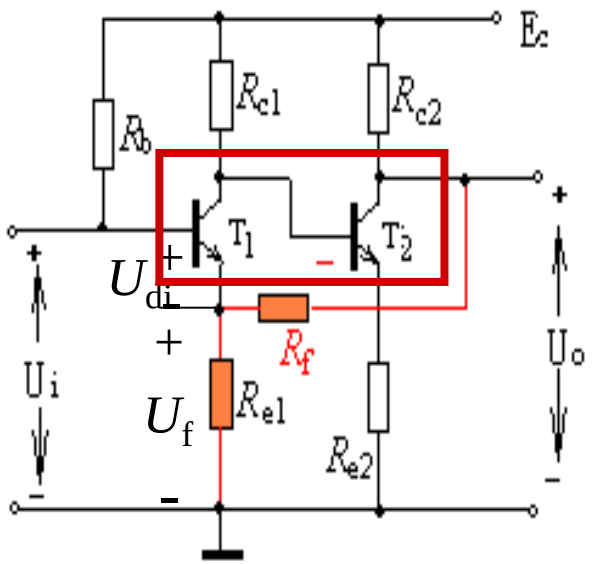
- No capacitors in the circuit , it's DC-AC mixed feedback

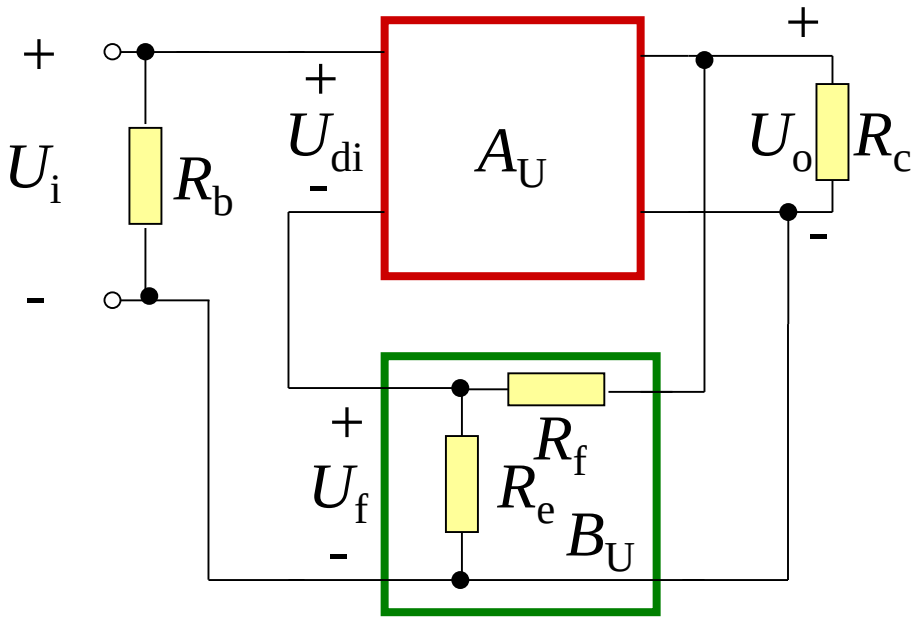


Overall, three feedbacks: two intra-stage current feedback and an inter-stage voltage feedback. All are DC-AC mixed series negative feedback

Voltage Series Negative Feedback

- **Block Diagram** -- based on AC path





Voltage Series Negative Feedback

Gain and Feedback Coefficient

open-loop gain

$$A_U = \frac{U_o}{U_{di}}$$

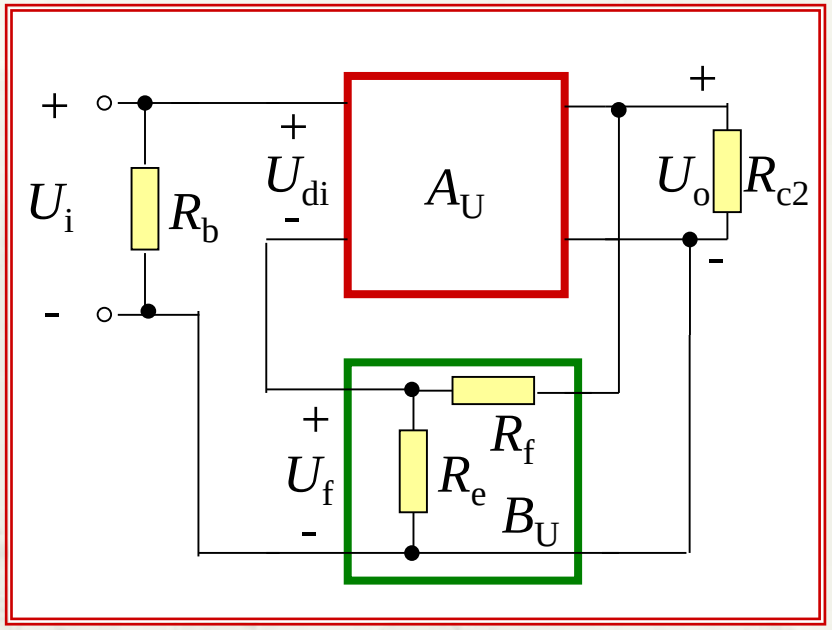
-- dimension less

closed-loop gain

$$A_{Uf} = \frac{U_o}{U_i}$$

feedback coefficient

$$B_U = \frac{U_f}{U_o}$$



feedback equation

$$A_{Uf} = \frac{A_U}{1 + A_U B_U}$$

feedback depth

$$F = 1 + A_U B_U$$

Current Parallel Negative Feedback

Determine the type of feedback

step 1: find feedback network

—R_{e2}, R_f/R_{e2}

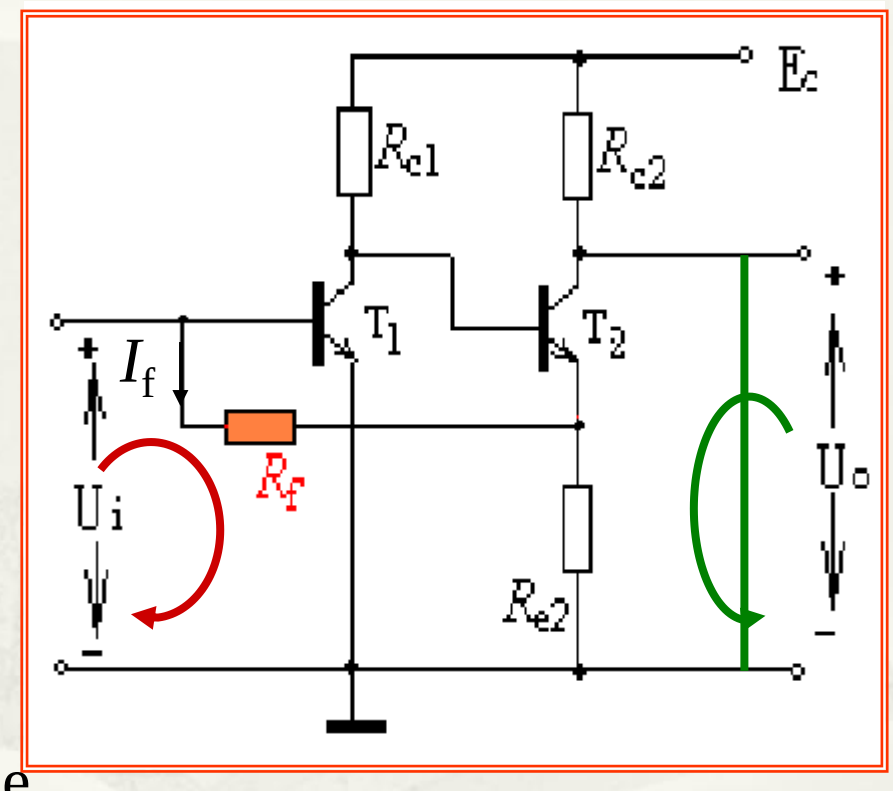
step 2: determine the type of feedback

① voltage or current **?**

- short the output voltage to ground
- feedback still exists. it's current feedback.

② series or parallel **?**

- input and feedback signals are added to the same terminal of the transistor. it's parallel feedback.



Current Parallel Negative Feedback

Determine the type of feedback

step 1: find feedback network

—R_{e2}, R_f/R_{e2}

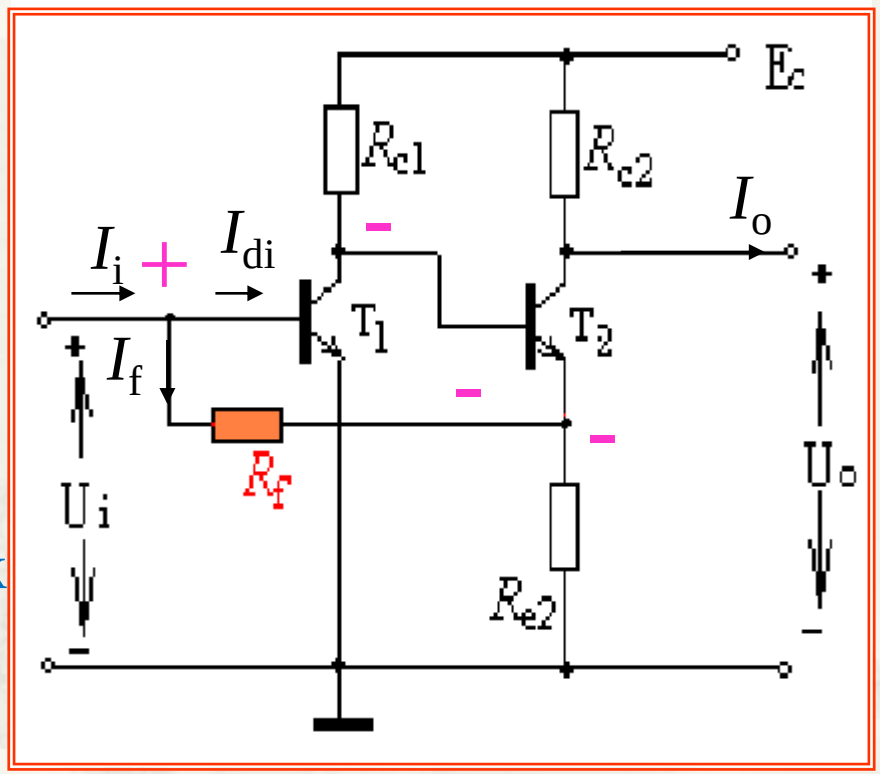
step 2: determine the type of feedback

③ positive or negative **?**

- Assume the polarity of input voltage is positive
- The polarity of feedback current is positive, it's negative feedback

④ DC or AC **?**

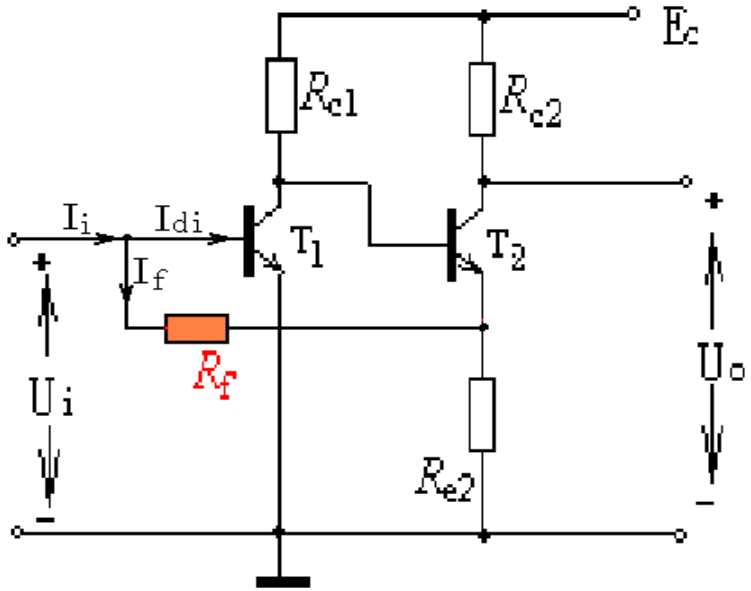
- No capacitors in the circuit , it's DC-AC mixed feedback

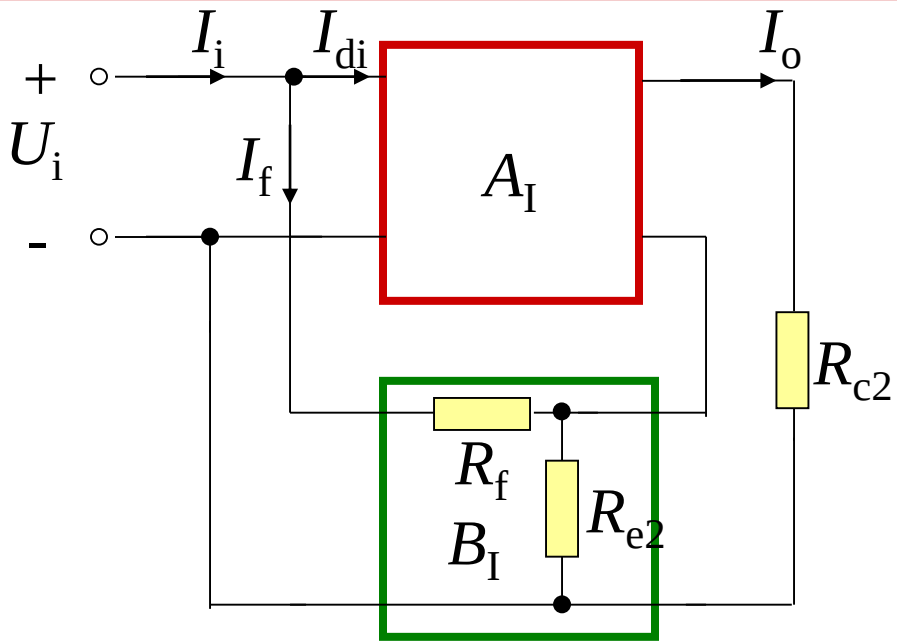


Overall, two feedbacks: one intra-stage series feedback and an inter-stage parallel feedback. Both are DC-AC current mixed negative feedback

Current Parallel Negative Feedback

- **Block Diagram** -- based on AC path





Current Parallel Negative Feedback

Gain and Feedback Coefficient

open-loop gain

$$A_I = \frac{I_o}{I_{di}}$$

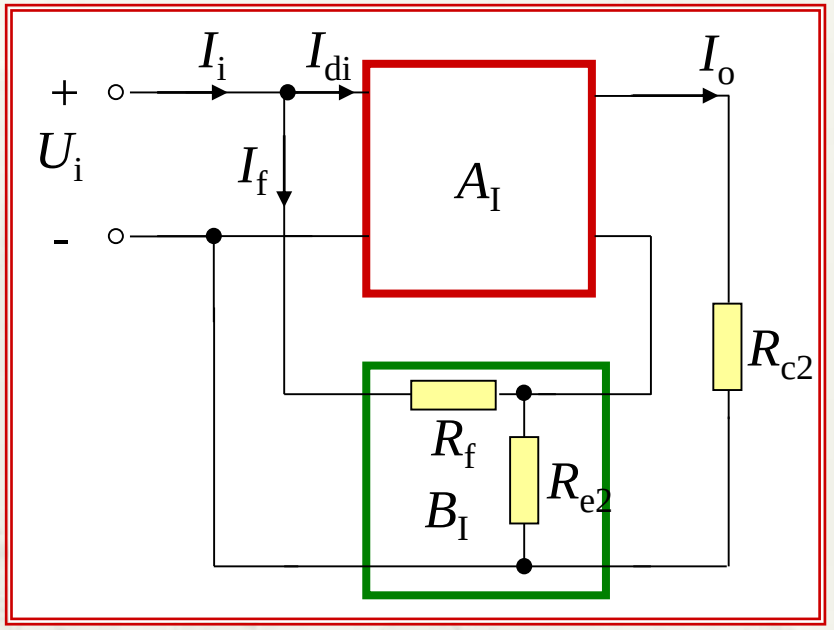
-- dimension less

closed-loop gain

$$A_{If} = \frac{I_o}{I_i}$$

feedback coefficient

$$B_I = \frac{I_f}{I_o}$$



feedback equation

$$A_{If} = \frac{A_I}{1 + A_I B_I}$$

feedback depth

$$F = 1 + A_I B_I$$

Voltage Parallel Negative Feedback

Determine the type of feedback

step 1: find feedback network — R_e , R_f

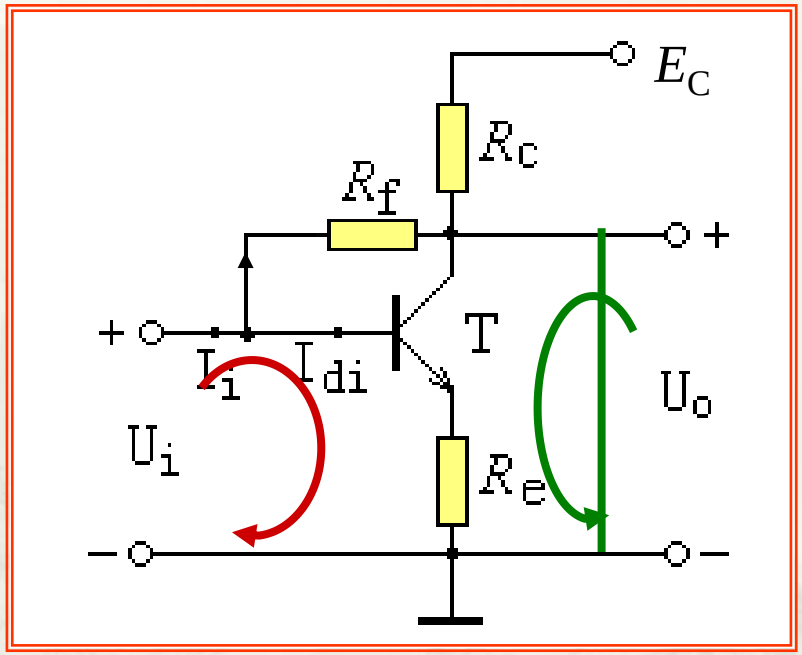
step 2: determine the type of feedback

① voltage or current **?**

- short the output voltage to ground
- feedback disappears. it's voltage feedback.

② series or parallel **?**

- input and feedback signals are added to the same terminal of the transistor. it's parallel feedback.



Voltage Parallel Negative Feedback

Determine the type of feedback

step 1: find feedback network — R_e , R_f

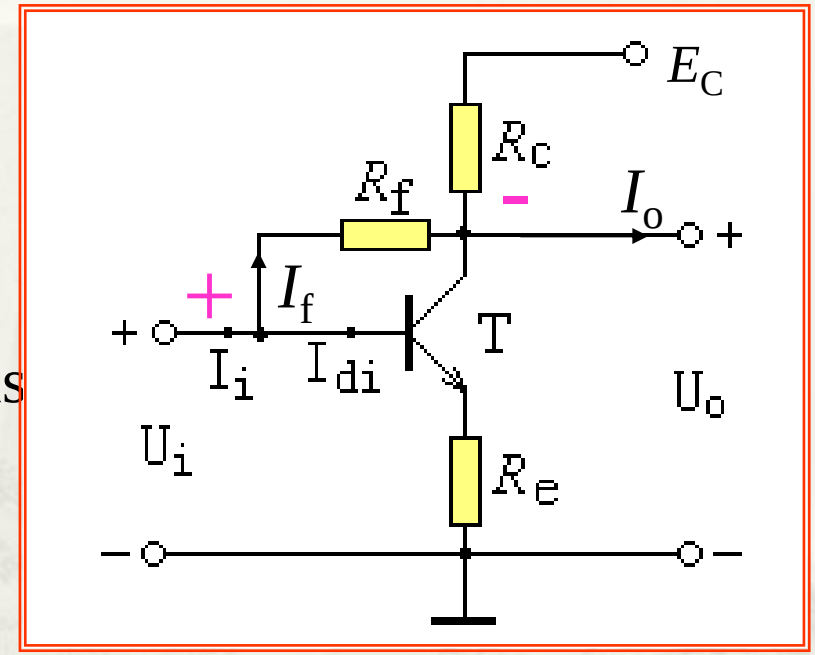
step 2: determine the type of feedback

③ positive or negative **?**

- Assume the polarity of input voltage is positive
- The polarity of feedback current is positive, it's negative feedback

④ DC or AC **?**

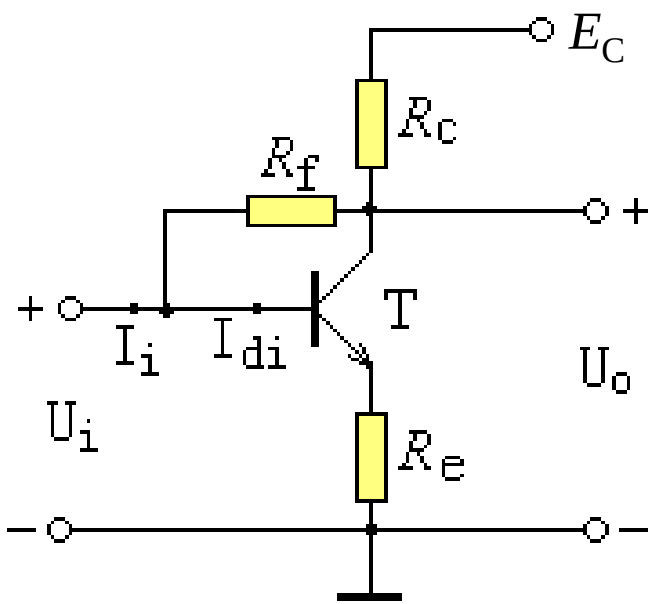
- No capacitors in the circuit , it's DC-AC mixed feedback

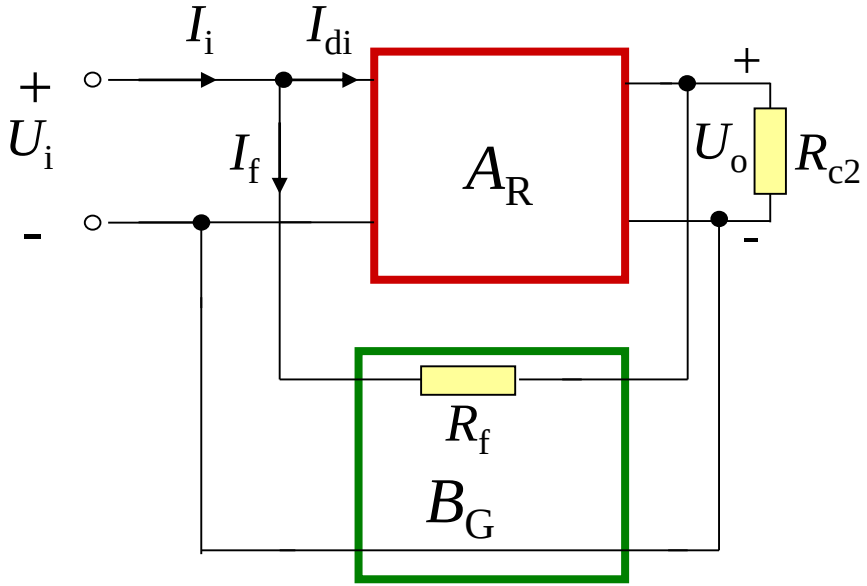


Overall, two feedbacks: a current series feedback and a voltage parallel feedback. Both are DC-AC mixed negative feedback

Voltage Parallel Negative Feedback

- **Block Diagram** -- based on AC path





Voltage Parallel Negative Feedback

Gain and Feedback Coefficient

open-loop gain

$$A_R = \frac{U_o}{I_{di}}$$

-- resistance dimension
-- transresistance gain

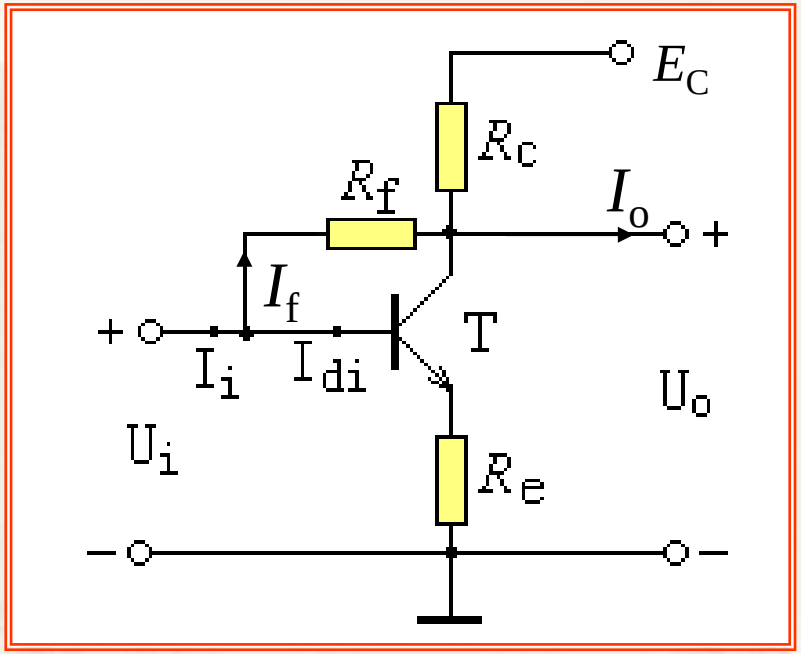
closed-loop gain

$$A_{Rf} = \frac{U_o}{I_i}$$

feedback coefficient

$$B_G = \frac{I_f}{U_o}$$

-- conductance dimension
-- transconductance coefficient



feedback equation

$$A_{Rf} = \frac{A_R}{1 + A_R B_G}$$

feedback depth

$$F = 1 + A_R B_G$$

5-5 Negative Feedback in Amplifiers

5-5-1 Basic Concepts

5-5-2 Category

5-5-3 Effects on Performance

5-5-4 Analysis Methods Based on Block Diagram

5-5-5 Closed-loop Stability of Multistage Amplifiers

5-5-6 Brief Summary

5-5-3 Effects on Performance

- Effect on Gain
- Effect on Bandwidth
- Effect on Non-linear Distortion
- Effect on Input and Output Resistances
- Effect of the Internal Resistance on the Performance of Negative Feedback AMPs

Effect on Gain

- **The gain is decreased by $|1+AB|$ times**

closed-loop gain: $A_f = \dfrac{A}{1+AB}$

No matter what kind of negative feedback, the AMP gain can be reduced by $|1+AB|$ times

If the open-loop gain changes ΔA from A to A'

$$\Delta A_f = \frac{A}{1+AB} - \frac{A'}{1+A'B} = \frac{A\text{-}A'}{(1+AB)(1+A'B)}$$

ΔA

$$\frac{\Delta A_f}{A_f} = \frac{1}{1+A'B} \cdot \frac{\Delta A}{A}$$

Effect on Gain

- **The stability of the gain is improved by $|1+A'B|$ times**

$$\frac{\Delta A_f}{A_f} = \frac{1}{1+A'B} \cdot \frac{\Delta A}{A}$$

The change of the closed-loop gain is $1/|1+A'B|$ of the change of the open-loop gain.

In deep negative feedback: $1+A'B>>1$

$$A_f = \frac{A'}{1+A'B} \approx \frac{1}{B}$$

The stability of the closed-loop gain is improved by $|1+A'B|$ times with negative feedback

Effect on Bandwidth

- **The passband is expanded by $|1+AB|$ times**

Bandwidth → $BW = f_h - f_l$

with negative feedback

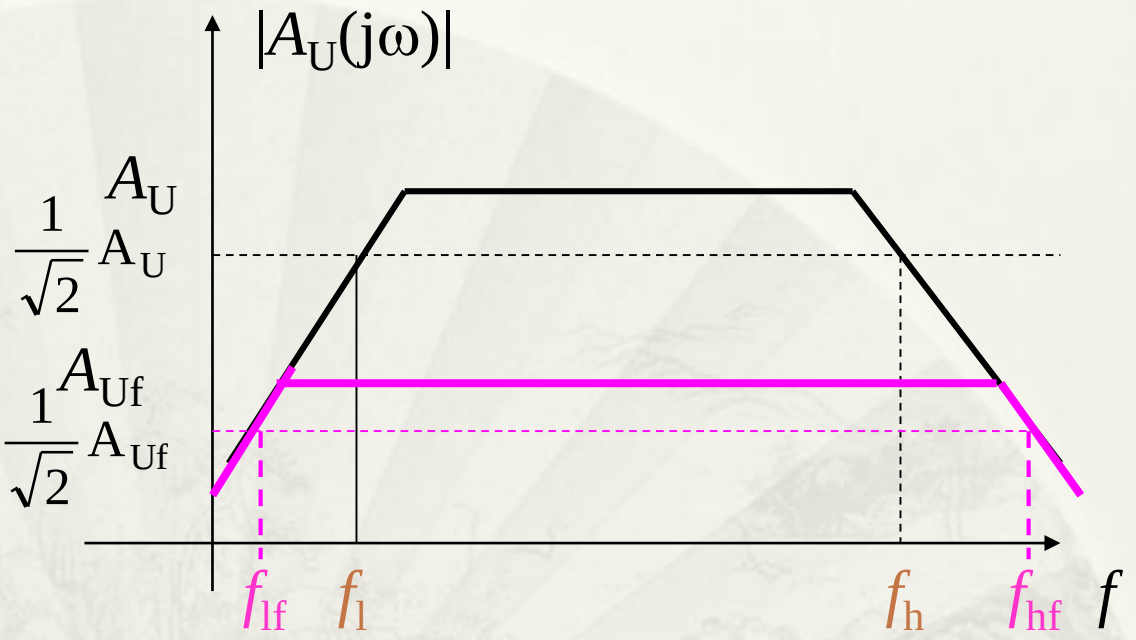
↓

gain is reduced

↓

passband is spread

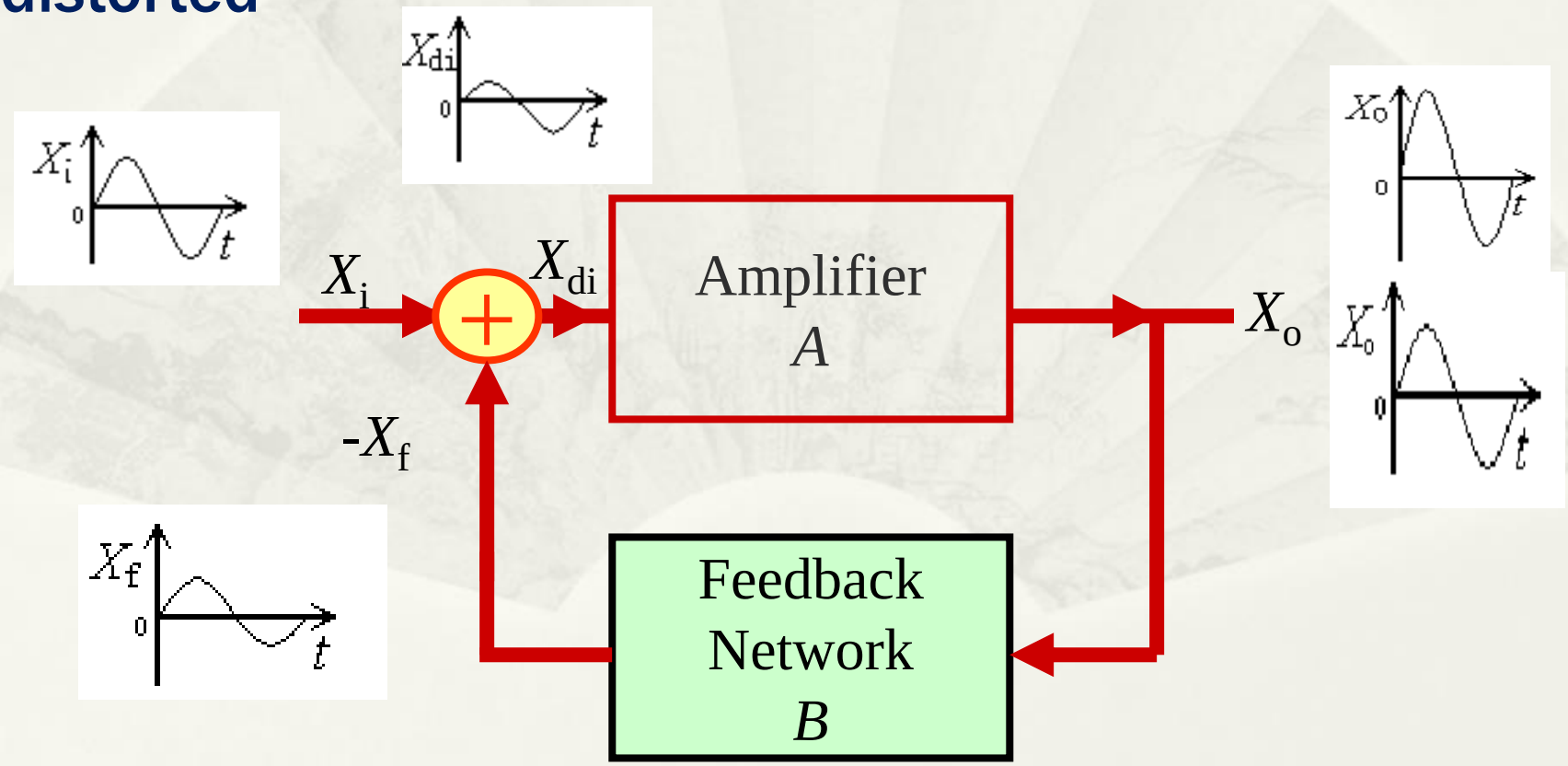
$BW_f = (1 + A_U B)\, BW$



The bandwidth is widen by $|1+AB|$ times with negative feedback

Effect on Non-linear Distortion

- The distortion with the loop can be reduced, but not disappear
- The distortion can not be reduced if the input itself is distorted



Effect on Input and Output Resistances

Effect on Input Resistance

- The effect depends on how to add the feedback at the input, series or parallel?
- The effect is not related to the output sampling

① **Series** negative feedback

no feedback: $r_i = U_{di}/I_i$

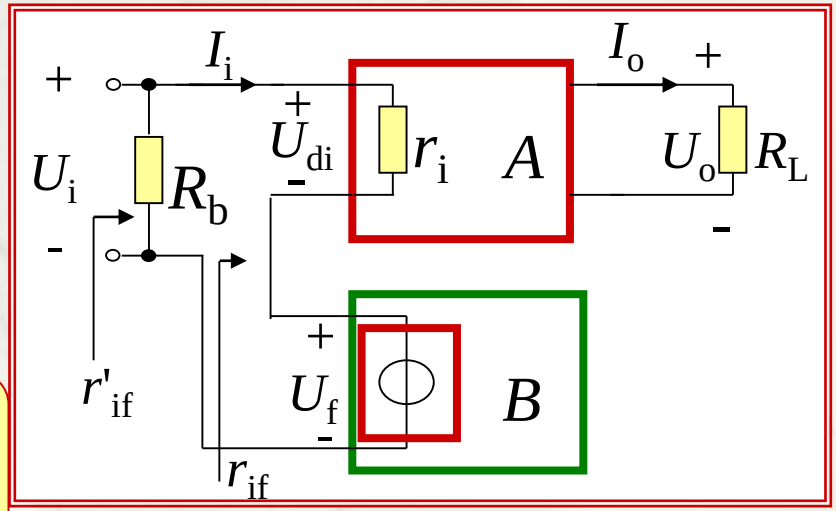
with feedback: $r_{if} = U_i/I_i$

For voltage series feedback

$$r_{if} = \frac{U_i}{I_i} = \frac{U_{di} + U_f}{I_i}$$

$U_f = B_U U_o$

$U_o = A_U U_{di}$



$$r_{if} = (1 + A_U B_U) r_i \qquad r'_{if} = R_b // r_{if}$$

Effect on Input and Output Resistances

➢ Effect on Input Resistance

① **Series** negative feedback

no feedback: $r_i = U_{di}/I_i$

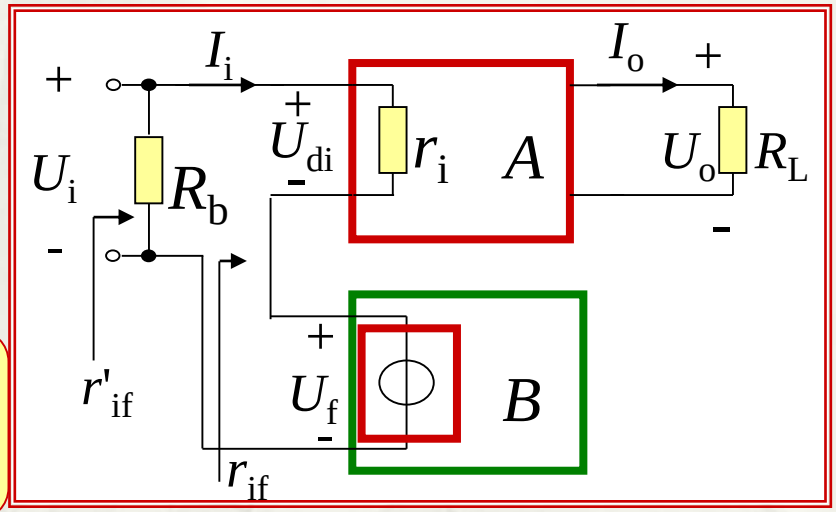
with feedback: $r_{if} = U_i/I_i$

For current series feedback

$$r_{if} = \frac{U_i}{I_i} = \frac{U_{di} + U_f}{I_i}$$

$U_f = B_R I_o$

$I_o = A_G U_{di}$



$$r_{if} = (1 + A_G B_R) r_i \qquad r'_{if} = R_b // r_{if}$$

Effect on Input and Output Resistances

➢ Effect on Input Resistance

① **Series** negative feedback

For voltage series feedback

$$r_{if} = (1 + A_U B_U) r_i \qquad r'_{if} = R_b // r_{if}$$

For current series feedback

$$r_{if} = (1 + A_G B_R) r_i \qquad r'_{if} = R_b // r_{if}$$

The input resistance is **increased** with **series** negative feedback
The increased multiple is equal to the **depth of feedback**

Effect on Input and Output Resistances

➢ Effect on Input Resistance

② **Parallel** negative feedback

no feedback: $r_i = U_i/I_{di}$ with feedback: $r_{if} = U_i/I_i$

For current parallel feedback

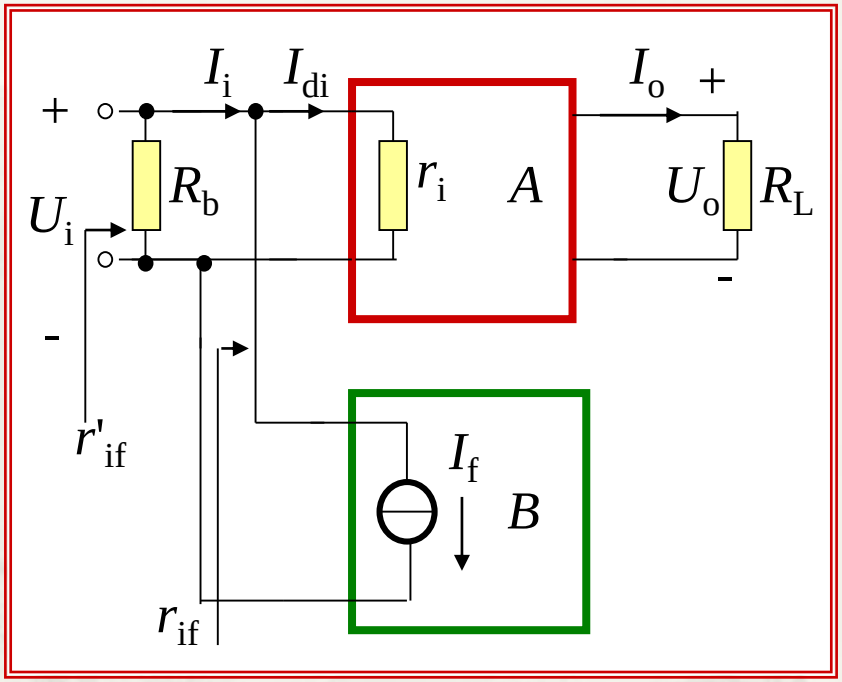
$$I_i = I_f + I_{di} \quad I_f = B_I I_o \quad I_o = A_I I_{di}$$

$$r_{if} = \frac{U_i}{I_i} = \frac{r_i}{1 + A_I B_I}$$

For voltage parallel feedback

$$r_{if} = \frac{r_i}{1 + A_R B_G}$$

$$r'_{if} = R_b // r_{if}$$



The input resistance is **decreased** with **parallel** negative feedback

The decreased multiple is equal to the **depth of feedback**

Effect on Input and Output Resistances

Effect on Input Resistance

- The effect only depends on how to add the feedback at the **input** (series or parallel), and is not related to the output sampling (current or voltage)
- The input resistance is **increased** with **series** negative feedback, and is **decreased** with **parallel** negative feedback.
- The increased or decreased multiple is equal to the **depth of feedback**.

Effect on Input and Output Resistances

Effect on Output Resistance

- The effect depends on the output sampling, voltage or current?
- The effect is not related to the connection at the input

Analysis method:

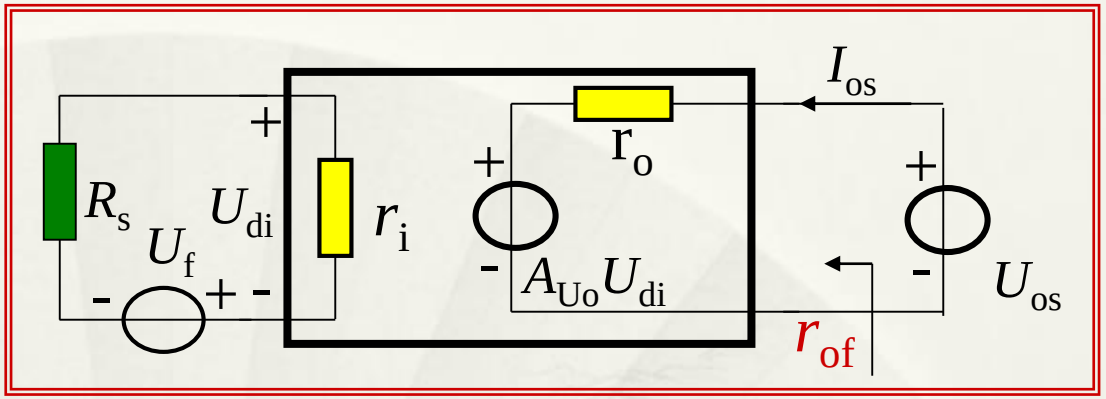
- Short the input voltage source or open the input current source. However, the internal resistor of the input source must be left.
- Move the load, and apply a voltage source U_{os} at the output. The generated current is I_{os}
- The output resistance is $r_{of} = U_{os}/U_{os}$
- Considering the effect of the load R_c in the collector, the output resistance is $r'_{of} = r_{of}//R_c$

Effect on Input and Output Resistances

➢ Effect on Output Resistance

① **Voltage** negative feedback

For voltage series feedback



open-loop voltage gain:

$$A_{Uo}=\left.\frac{U_o}{U_{di}}\right|_{R'_L=\infty}$$

$$A_{Uso}=\frac{r_i}{R_s+r_i}A_{Uo} \qquad U_{di}=\frac{-r_i}{R_s+r_i}U_f \qquad U_f=B_U U_{os}$$

$$I_{os}=\frac{U_{os}-A_{Uo}U_{di}}{r_o}=\frac{U_{os}+A_{Uso}U_f}{r_o}=\frac{U_{os}+A_{Uso}B_U U_{os}}{r_o}=\frac{U_{os}\left(1+A_{Uso}B_U\right)}{r_o}$$

$$r_{of}=\frac{U_{os}}{I_{os}}=\frac{r_o}{1+A_{Uso}B_U}$$

$$r'_{of}=r_{of}//R_c$$

The output resistance is **decreased**

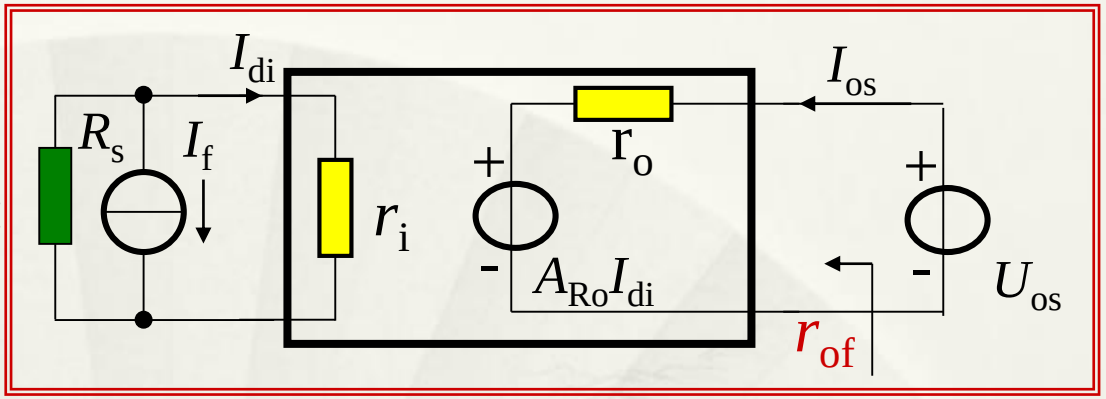
Effect on Input and Output Resistances

Effect on Output Resistance

① **Voltage** negative feedback

For voltage parallel feedback

open-loop transresistance gain:



$$A_{Ro}=\left.\frac{U_o}{I_{di}}\right|_{R'_L=\infty}$$

$$A_{Rso}=\frac{R_s}{R_s+r_i}A_{Ro} \qquad I_{di}=\frac{-R_s}{R_s+r_i}I_f \qquad I_f=B_G U_{os}$$

$$U_{os}=I_{os}r_o+A_{Ro}I_{di}=I_{os}r_o-B_G A_{Rso}U_{os} \Rightarrow U_{os}=\frac{r_o}{1+A_{Rso}B_G}I_{os}$$

$$r_{of}=\frac{U_{os}}{I_{os}}=\frac{r_o}{1+A_{Rso}B_G}$$

$$r'_{of}=r_{of}//R_c$$

The output resistance is **decreased**

Effect on Input and Output Resistances

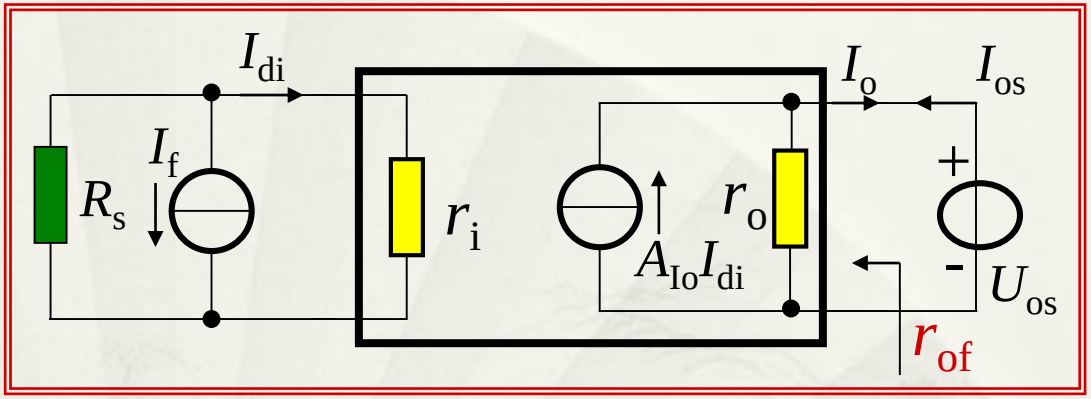
Effect on Output Resistance

② **Current** negative feedback

For current parallel feedback

short-output current gain:

$$A_{Io}=\left.\frac{I_o}{I_{di}}\right|_{R'_L=0}$$



$$A_{Iso}=\frac{R_s}{R_s+r_i}A_{Io} \qquad I_{di}=\frac{-R_s}{R_s+r_i}I_f \qquad I_f=B_I I_o \qquad I_o=-I_{os}$$

$$I_o=A_{Io}I_{di}-\frac{U_{os}}{r_o}=-A_{Iso}B_I I_o-\frac{U_{os}}{r_o} \Rightarrow U_{os}=\left(1+A_{Iso}B_I\right)r_o I_{os}$$

$$r_{of}=\frac{U_{os}}{I_{os}}=(1+A_{Iso}B_I)r_o \qquad r'_{of}=r_{of}//R_c$$

The output resistance is **increased**

Effect on Input and Output Resistances

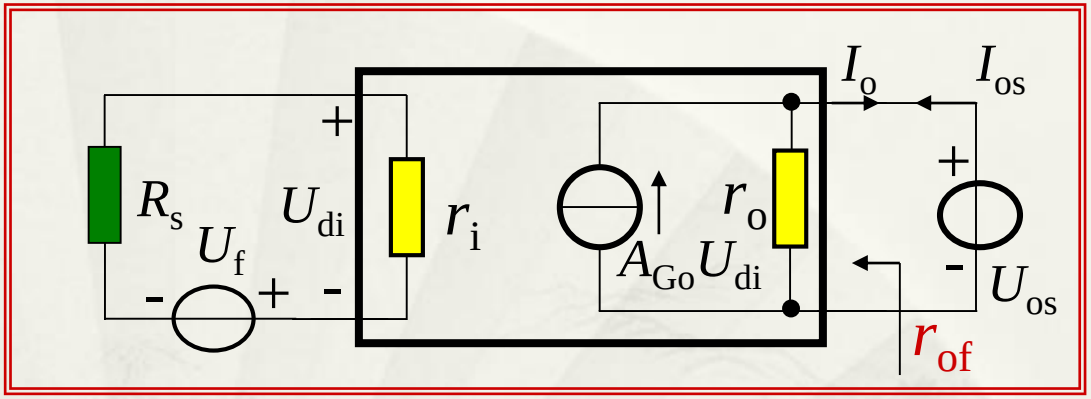
Effect on Output Resistance

② **Current** negative feedback

For current series feedback

short-output transconductance gain:

$$A_{Go}=\left.\frac{I_o}{U_{di}}\right|_{R'_L=0}$$



$$A_{Gso}=\frac{r_i}{R_s+r_i}A_{Go} \qquad U_{di}=\frac{-r_i}{R_s+r_i}U_f \qquad U_f=B_R I_o \qquad I_o=-I_{os}$$

$$I_o=A_{Go}U_{di}-\frac{U_{os}}{r_o}=-A_{Gso}B_R I_o-\frac{U_{os}}{r_o} \Rightarrow U_{os}=(1+A_{Gso}B_R)r_o I_{os}$$

$$r_{of}=\frac{U_{os}}{I_{os}}=(1+A_{Gso}B_R)r_o \qquad r'_{of}=r_{of}//R_c$$

The output resistance is **increased**

Effect on Input and Output Resistances

Effect on Output Resistance

- The effect only depends on the **output** sampling (voltage or current), and is not related to the connection at input (current or voltage)
- The output resistance can be **decreased** with **voltage** negative feedback, and is **increased** with **current** negative feedback.
- The increased or decreased multiple is equal to the **depth of feedback**.
- **Voltage** negative feedback can stabilize the output **voltage**, and **current** negative feedback can stabilize the output **current**.

Effect on Input and Output Resistances

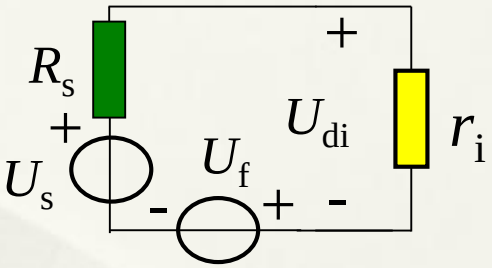
- **Summary**

	voltage series	current series	voltage parallel	current parallel
input resistance	increase	increase	decrease	decrease
output resistance	decrease	increase	decrease	increase
stability of output	voltage	current	voltage	current

Effect of the Internal Resistance on the Performance of Negative Feedback AMPs

Series Negative Feedback

- voltage source U_s should be applied
- the effect of R_s is on the basic amplifier



Example: voltage series feedback

$$A_{Us} = \frac{r_i}{R_s + r_i} A_U$$

$$A_{Uf} = \frac{A_U}{1 + A_U B_U}$$

$$\Rightarrow \quad A_{Ufs} = \frac{A_{Us}}{1 + A_{Us} B_U} \qquad F = 1 + A_{Us} B_U$$

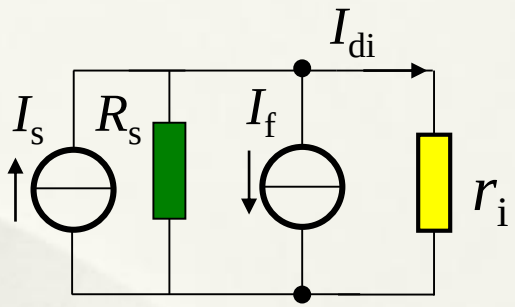
When $R_s \to \infty$, $A_{Us} \to 0$, $A_{Ufs} = A_{Us}$ → negative feedback doesn't work

For **series** negative feedback, the internal resistance of the voltage source should be **low** to achieve better feedback.

Effect of the Internal Resistance on the Performance of Negative Feedback AMPs

Parallel Negative Feedback

- current source I_s should be applied
- the effect of R_s is on the basic amplifier



Example: current parallel feedback

$$A_{Is} = \frac{R_s}{R_s + r_i} A_I$$

$$A_{If} = \frac{A_I}{1 + A_I B_I}$$

$$\Rightarrow \quad A_{Ifs} = \frac{A_{Is}}{1 + A_{Is} B_I}$$

$$F = 1 + A_{Is} B_I$$

When $R_s \to 0$, $A_{Is} \to 0$, $A_{Ifs} = A_{Is}$ → negative feedback doesn't work

For **parallel** negative feedback, the internal resistance of the current source should be **high** to achieve better feedback.

5-5 Negative Feedback in Amplifiers

5-5-1 Basic Concepts

5-5-2 Category

5-5-3 Effects on Performance

5-5-4 Analysis Methods Based on Block Diagram

5-5-5 Closed-loop Stability of Multistage Amplifiers

5-5-6 Brief Summary

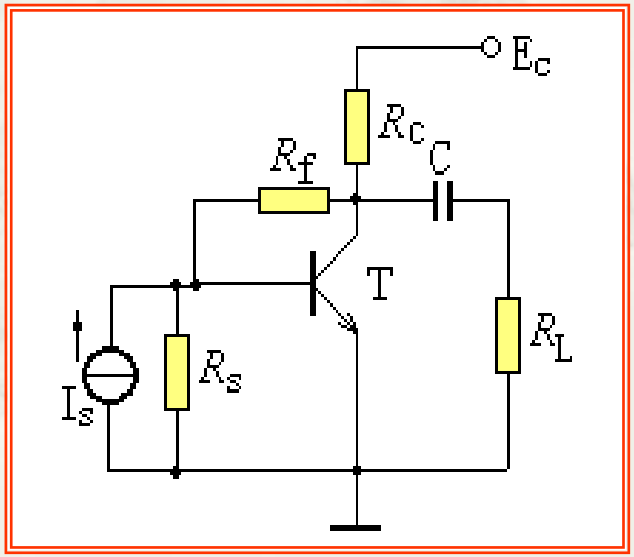
5-5-4 Analysis Method Based on Block Diagram

- Analysis Method Based on Block Diagram
- Examples

Analysis Method Based on Block Diagram

Analysis Flow

- Divide the feedback AMP into two parts: basic AMP and feedback network.
- Calculate the gain A of the basic AMP and feedback coefficient B of the feedback network, respectively.
- Calculate the gain A_{uf} of the feedback AMP.
- **Note:** the load effect of the feedback network must be considered when gain A is calculated.



Analysis Method Based on Block Diagram

Analysis Flow

Determine the type of feedback

Determine the output
voltage feedback: $X_o=U_o$
current feedback: $X_o=I_o$

Determine the input & feedback
series feedback: X_i, X_f are voltage
parallel feedback: X_i, X_f are current

↓

Draw the basic AMP
Consider the load effects of the feedback network

↓

Calculate the parameters of the basic AMP -- A, r_i, r_o

↓

Calculate the parameters of the feedback AMP -- B, F, A_f , r_{if}, r_{of}, r'_{if}, r'_{of}

↓

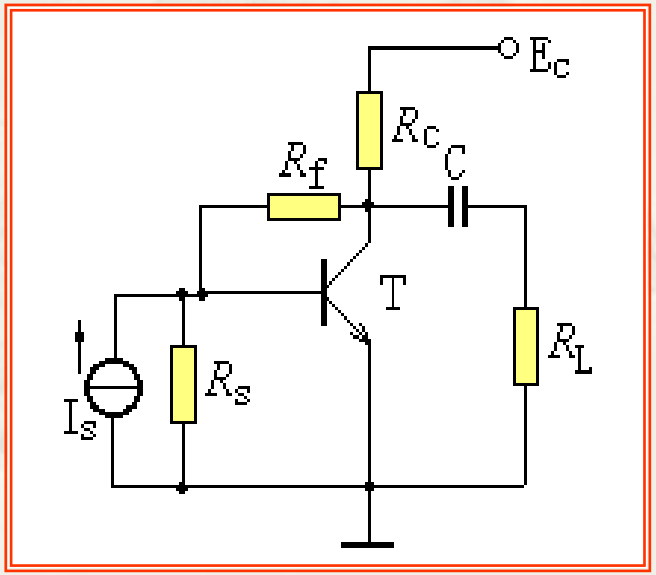
calculate other parameters

Analysis Method Based on Block Diagram

Rule of the basic AMP circuit

Consider the feedback effects on input and output:

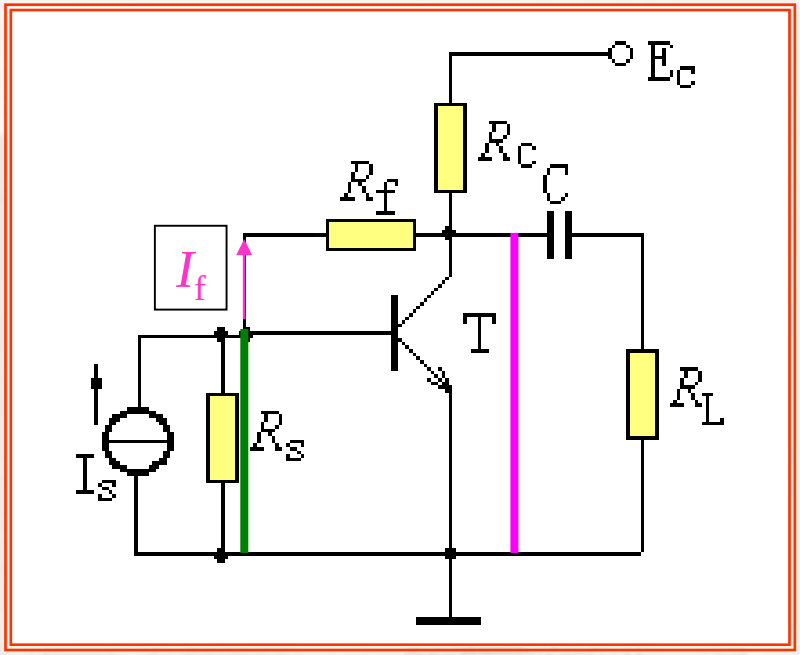
- Determine the input circuit
  - ✓ for voltage feedback, let U_o=0, short the output.
  - ✓ for current feedback, let I_o=0, open the output.
- Determine the output circuit
  - ✓ for parallel feedback, let U_i=0, short the input.
  - ✓ for series feedback, let I_i=0, open the input.

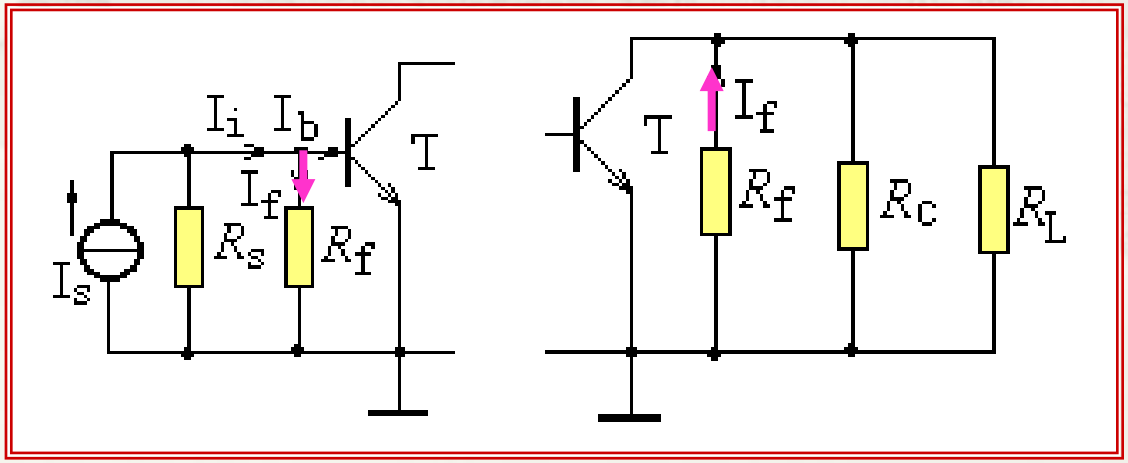


Examples

Voltage Parallel Feedback AMP

1. Determine the type of feedback
   - ✓ feedback component: R_f
   - ✓ feedback type:
     voltage parallel negative feedback
2. Draw the basic AMP circuit
   - ✓ feedback load effect on input loop
   - ✓ feedback load effect on output loop

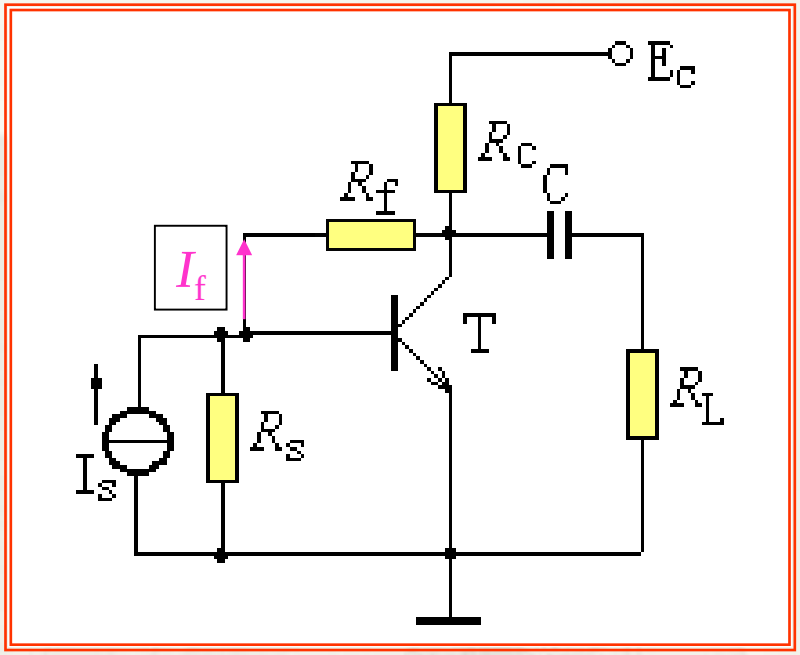


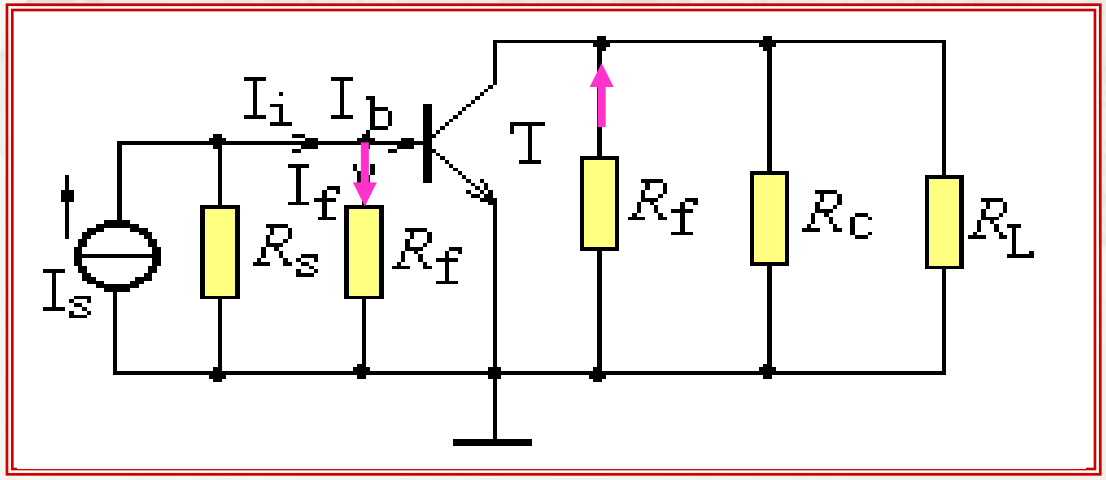


Examples

Voltage Parallel Feedback AMP

1. Determine the type of feedback
   - ✓ feedback component: R_f
   - ✓ feedback type:
     voltage parallel negative feedback
2. Draw the basic AMP circuit
   - ✓ feedback load effect on input loop
   - ✓ feedback load effect on output loop

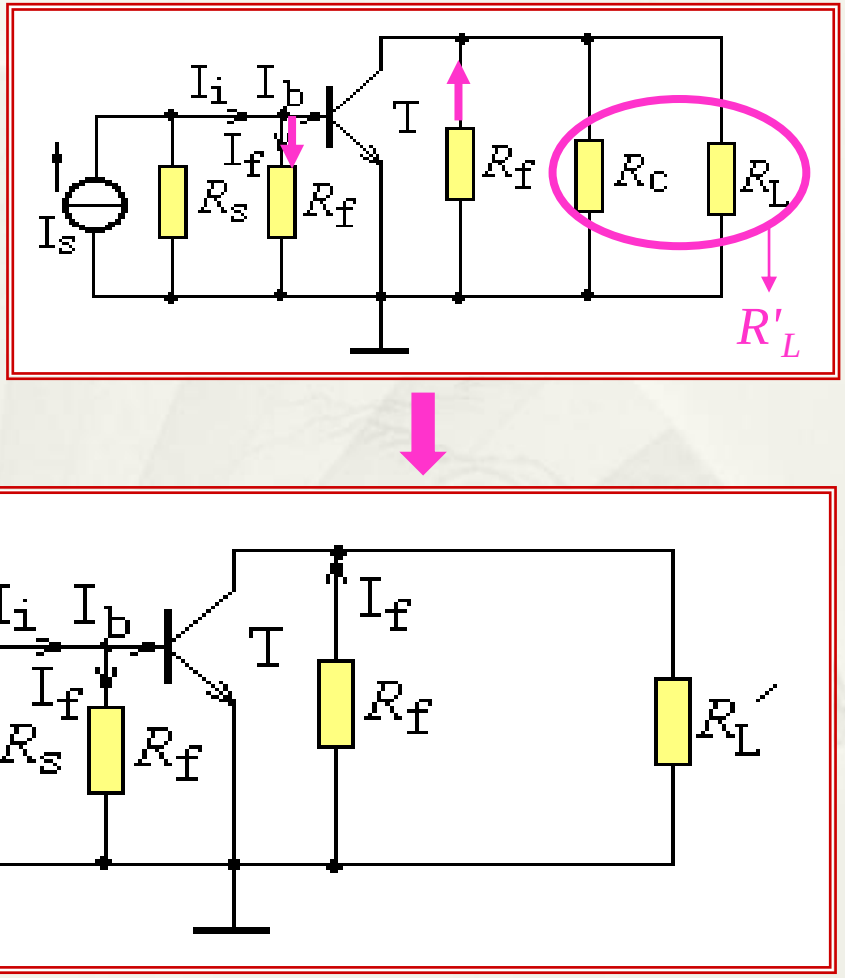




Examples

- **Voltage Parallel Feedback AMP**

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit
   - ✓ simplify the circuit



Examples

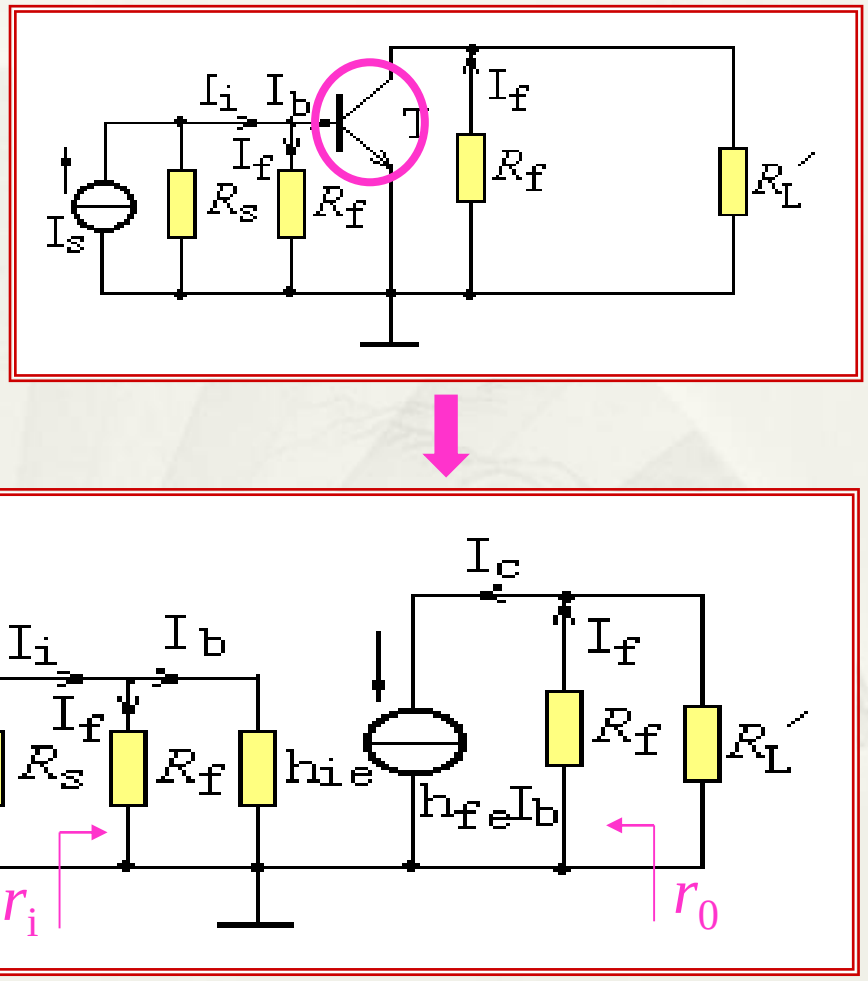
Voltage Parallel Feedback AMP

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit
   ① simplify the circuit
   ② replace the BJT to its model
   --small signal model
   ③ input resistance

$$r_i = R_f // h_{ie}$$

   ④ output resistance

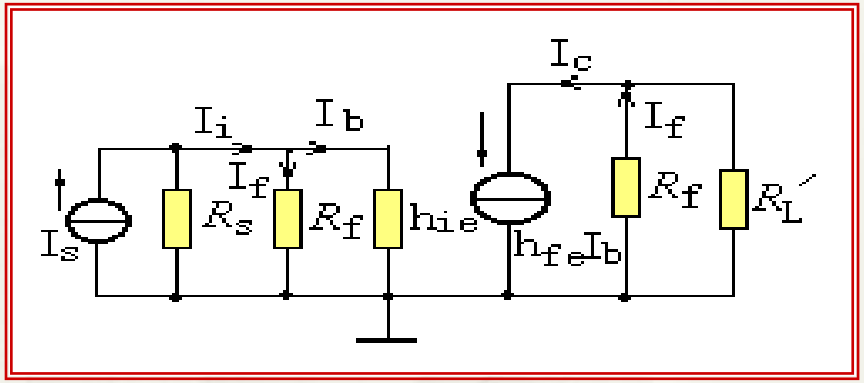
$$r_o = R_f$$



Examples

Voltage Parallel Feedback AMP

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit
   ⑤ open-loop transresistance gain



$$A_R = \frac{U_o}{I_i} = \frac{-I_c(R_f // R'_L)}{I_i} = \frac{-h_{fe} I_b (R_f // R'_L)}{I_i}$$

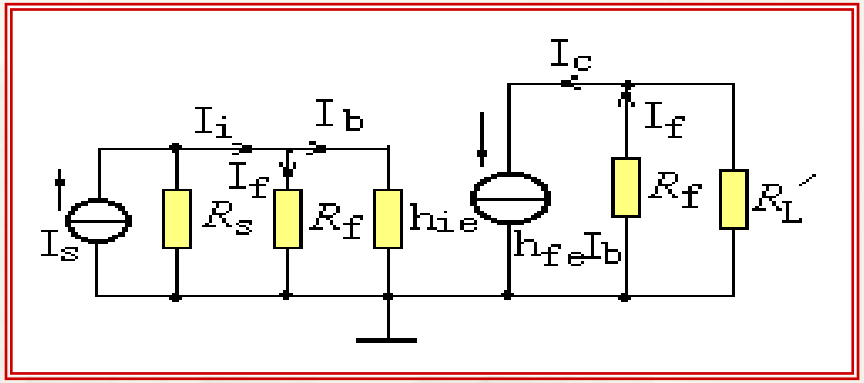
$$I_b = \frac{R_f}{R_f + h_{ie}} I_i \longrightarrow A_R = \frac{-h_{fe} R_f (R_f // R'_L)}{R_f + h_{ie}}$$

consider the effect of R_S: $A_{Rs} = \frac{R_s}{R_s + r_i} A_R$

Examples

Voltage Parallel Feedback AMP

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit
4. Calculate the feedback AMP



$$(1)\ B_G = \frac{I_f}{U_o} = \frac{I_f}{-I_f R_f} = -\frac{1}{R_f}$$

$$(2)\ F = 1 + A_R B_G$$

$$(3)\ r_{if} = r_i / F$$

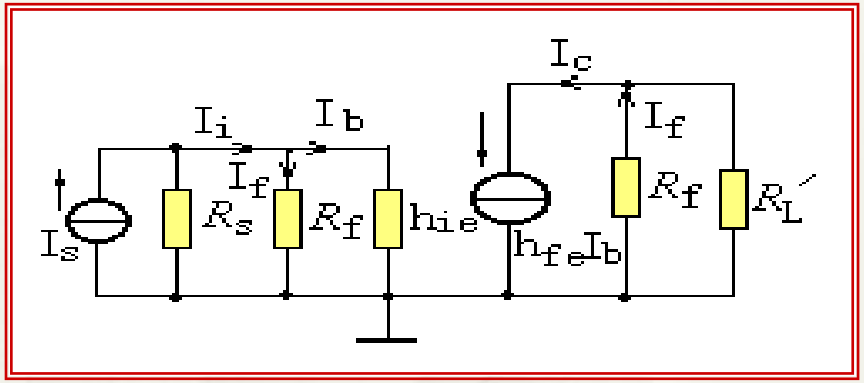
$$(4)\ A_{Rf} = A_R / F$$

parallel feedback reduces the input resistance by F times

Examples

Voltage Parallel Feedback AMP

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit
4. Calculate the feedback AMP



⑤ consider the effect of R_S:

$$A_{Rsf} = \frac{A_{Rs}}{1 + A_{Rs} B_G}$$

⑥ $$A_{Usf} = \frac{U_o}{U_s} = \frac{U_o}{I_s R_s} = \frac{A_{Rsf}}{R_s}$$

In deep feedback: $AB >> 1$

$$\Rightarrow A_{Rsf} \approx \frac{1}{B_G} = -R_f \quad A_{Usf} \approx -R_f / R_s$$

⑦ output resistance

$$r_{of} = \frac{r_o}{1 + A_{Rso} B_G}$$

$$A_{Rso|R'_L=\infty} = \frac{R_s}{R_s + r_i} A_{R|R'_L=\infty}$$

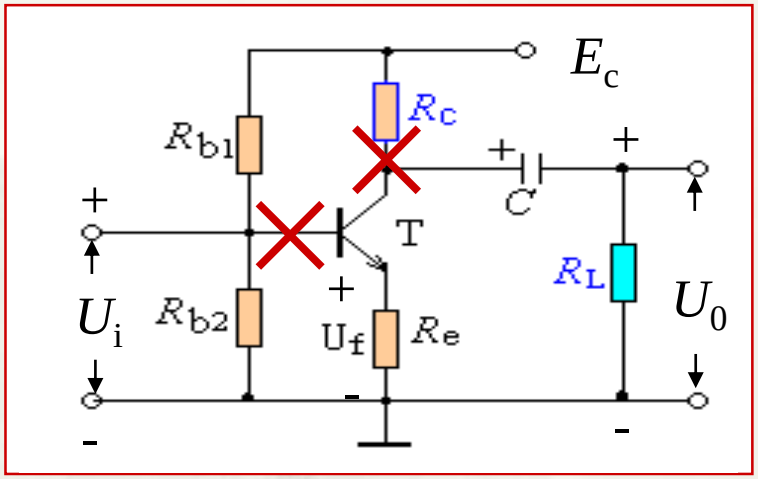
consider the effect of R_c:

$$r'_{of} = r_{of} // R_c$$

Examples

Current Series Feedback AMP

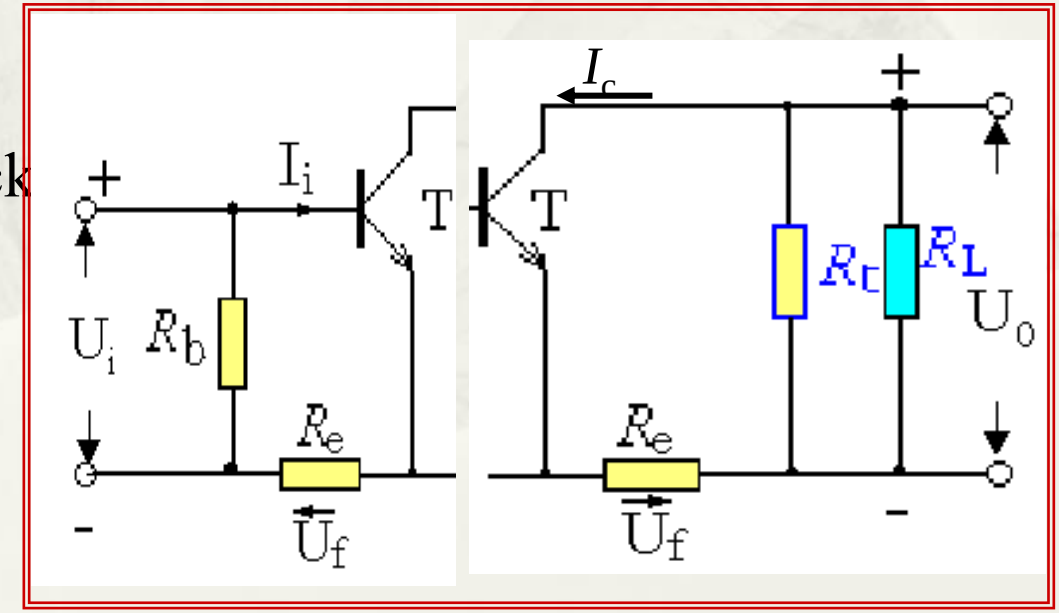
The parameters of the circuit are given.
h_{fe}=100, h_{ie}=1kΩ, $h_{re}\approx h_{oe}\approx$0, $R_s\approx$0,
R_e=100Ω, R_{b1}=100kΩ, R_{b2}=50kΩ,
R_c=2kΩ, R_L=2kΩ.
Determine A_{Gf}, r_{if}, r_{of} and A_{uf}



Solution:

1. Determine the type of feedback
   - ✓ feedback component: R_e
   - ✓ feedback type:
   current series negative feedback

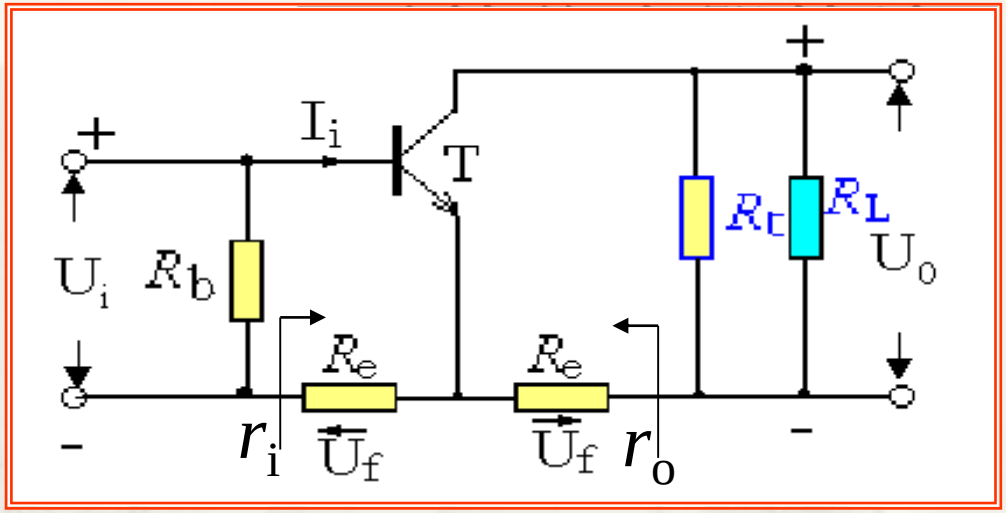
2. Draw the basic AMP circuit
   - ✓ feedback load effect on input loop
   - ✓ feedback load effect on output loop



Examples

Current Series Feedback AMP

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit



$r_i = h_{ie} + R_e = 1.1\text{k}\Omega$

$r_o = 1/h_{oe} + R_e \to \infty$

$$A_G = \frac{I_o}{U_i} = \frac{h_{fe} I_i}{I_i r_i} = \frac{h_{fe}}{r_i} = \frac{100}{1.1} = 90.9\text{mA/V}$$

4. Calculate the feedback AMP

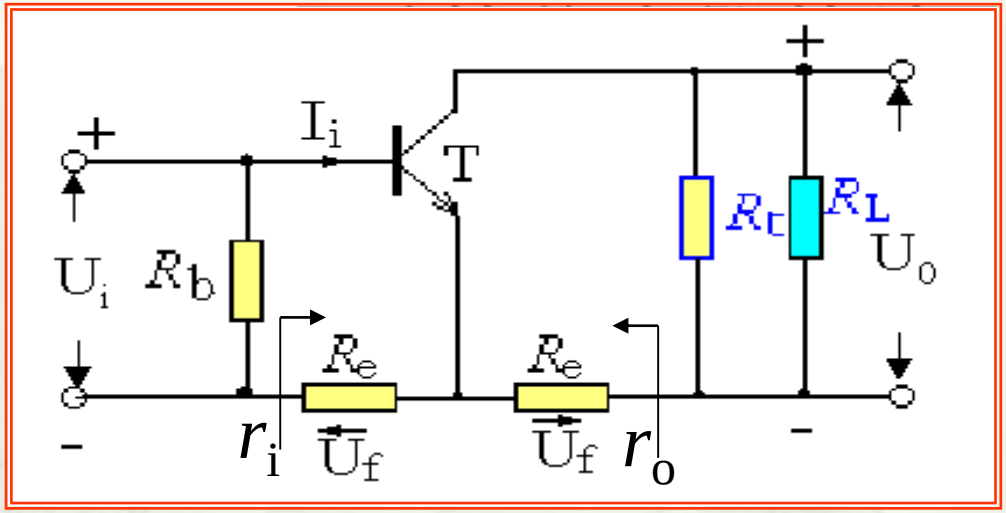
$$B_R = \frac{U_f}{I_o} = \frac{I_o R_e}{I_o} = R_e = 0.1\text{K}\Omega$$

$$F = 1 + A_G B_R = 1 + 90.9 \times 0.1 = 10.1$$

Examples

Current Series Feedback AMP

1. Determine the type of feedback
2. Draw the basic AMP circuit
3. Calculate the basic AMP circuit
4. Calculate the feedback AMP



$$A_{Gf} = \frac{A_G}{1 + A_G B_R} = \frac{90.9}{10.1} = 9.09\text{mA/V}$$

$$r_{if} = r_i \cdot F = 1.1 \times 10.1 = 11.1\text{k}\Omega$$

$$r'_{if} = r_{if} // R_{b1} // R_{b2} = 11.1 // 50 // 100 = 8.33\text{k}\Omega$$

$$r_{of} = (1 + A_{Gso} B_R) r_o \to \infty$$

$$r'_{of} = r_{of} // R_c \approx R_c = 2\text{k}\Omega$$

$$A_{Uf} = \frac{U_o}{U_i} = \frac{-I_o R'_L}{U_i} = -A_{Gf} R'_L = -9 \times (2 // 2) = -9$$

5-5 Negative Feedback in Amplifiers

5-5-1 Basic Concepts

5-5-2 Category

5-5-3 Effects on Performance

5-5-4 Analysis Methods Based on Block Diagram

5-5-5 Closed-loop Stability of Multistage Amplifiers

5-5-6 Brief Summary

5-5-4 Analysis Method Based on Block Diagram

- Self-Oscillation of Multipole Closed-Loop System
- Determination of Closed-Loop Stability

Self-Oscillation of Multipole Closed-Loop System

Self-Oscillation

- ✓ negative can improve the performance of AMPs.
- ✓ improper application of negative feedback will cause self-oscillation of AMPs

Definition:

When U_i=0, there still is a signal with a certain frequency and magnitude at output terminal.

Reason:

- ✓In IF band, the AMP has a phase shift of 180°
- ✓In other bands,
  - the AMP has extra phase shift ϕ
  - When ϕ=180°, the total phase shift is 360°. Negative feedback changes to positive feedback. Self-oscillation is generated.

Self-Oscillation of Multipole Closed-Loop System

Condition of Self-Oscillation

For negative feedback system $A_f = \dfrac{A}{1+AB}$

When $AB = -1$, $A_f \to \infty$. It means that there is a signal at the output. The AMP is self-oscillated.

Condition: $AB = -1$

magnitude: $|AB| = 1$

phase: $\arg A(j\omega)B(j\omega) = \pm 180^0$

Starting oscillation condition: $|AB| > 1$

When the AMP switches on, the output signal has a process from small to large to stable.

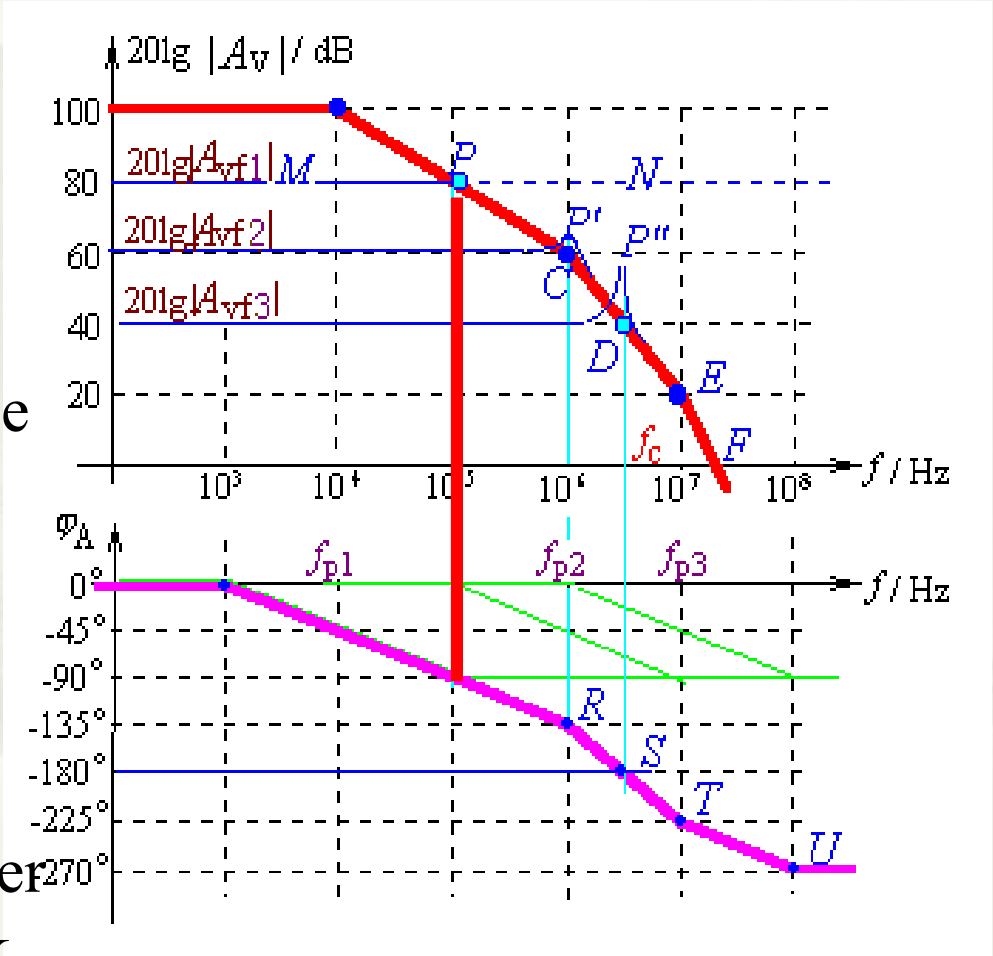
Self-Oscillation of Multipole Closed-Loop System

Condition of Self-Oscillation

Example: 3 feedback AMPs

For A_{vf1}

- ✓ the corresponding additional phase shift at P is $\varphi_A = -90°$, which doesn't satisfy the phase condition.
- ✓ The feedback AMP is stable, and oscillation doesn't occur.
- ✓ However, the feedback has basically the same vector direction as the input, and enter the range of positive feedback.



- ✓ the gain is increased when the signal frequency is close to 10^6Hz, P'

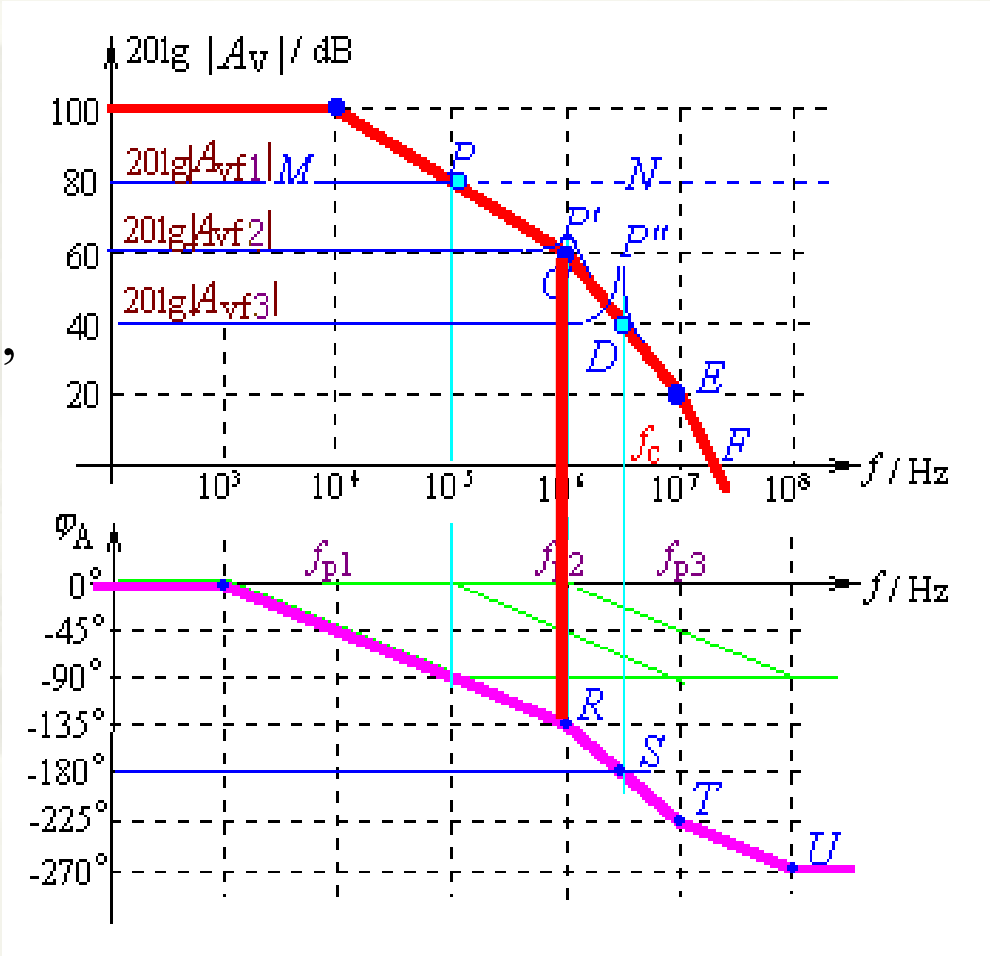
Self-Oscillation of Multipole Closed-Loop System

Condition of Self-Oscillation

Example: 3 feedback AMPs

For A_{vf2}

- ✓ the corresponding additional phase shift at P' is $\varphi_A = -135°$, which doesn't satisfy the phase condition.
- ✓ The feedback AMP is stable, and oscillation doesn't occur.



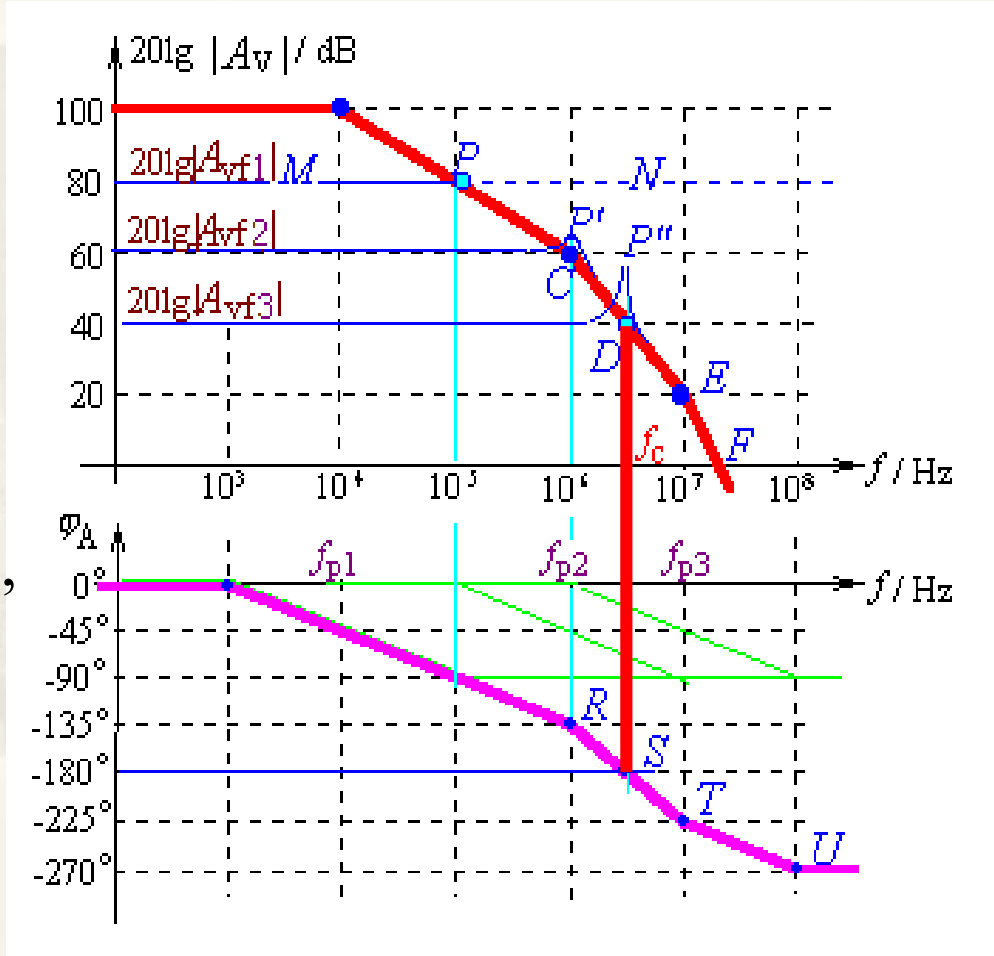
Self-Oscillation of Multipole Closed-Loop System

Condition of Self-Oscillation

Example: 3 feedback AMPs

For A_{vf3}

- ✓ At P", $|AF|=100\times10^{-2}=1$, which satisfies the magnitude condition.
- ✓ the corresponding additional phase shift at P" is $\varphi_A = -180°$, which satisfies the phase condition.
- ✓ self-oscillation occurs.



Determination of Closed-Loop Stability

Determination Based on Loop Gain

Change the Y coordinate of bode plot from $20\lg A$ to $20\lg AB$ for analyzing self-oscillation

$$20\lg|AB|=20\lg|A|+20\lg|B|=20\lg|A|-20\lg|1/B|$$

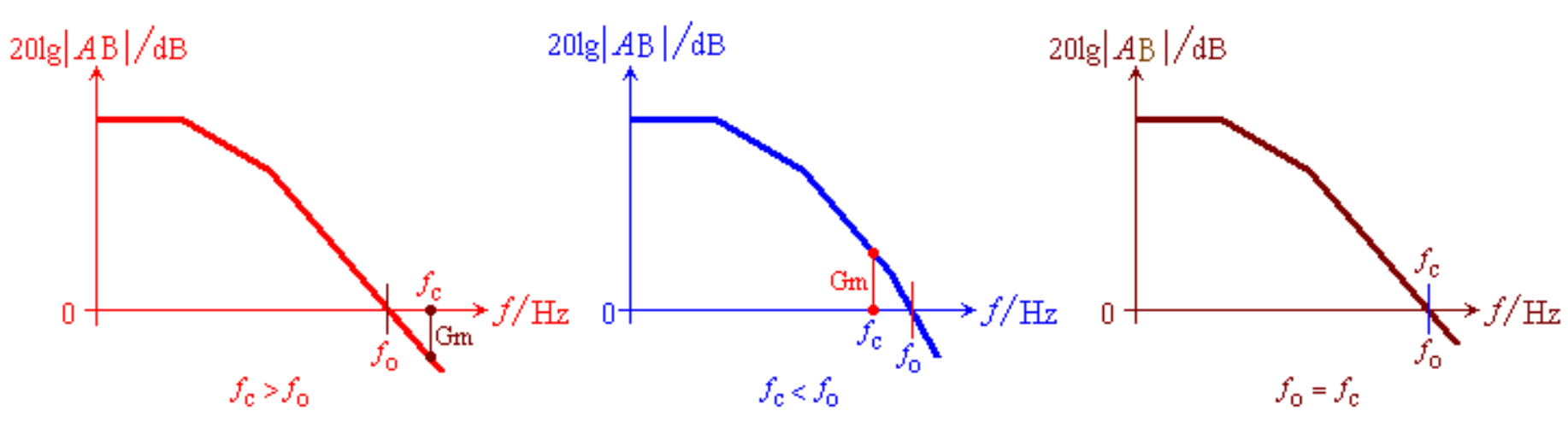
For magnitude condition: $|AB|=1$ ⇓

$$20\lg|AB| = 20\lg|A|-20\lg|1/B| = 0\ \text{dB}$$

So, the bode plot with $20\lg AB$ coordinate is the bode plot with $20\lg A$ coordinate subtracting $20\lg|1/B|$.

Determination of Closed-Loop Stability

Determination Based on Loop Gain



(a) $f_c > f_0$, $G_m < 0$ (b) $f_c < f_0$,$G_m > 0$ (c) $f_c = f_0$, $G_m = 0$

f_0 — gain crossover frequency: satisfy $|AB| = 1 = 0\text{dB}$

f_c — phase crossover frequency: satisfy $\phi = \pm 180°$

magnitude margin:

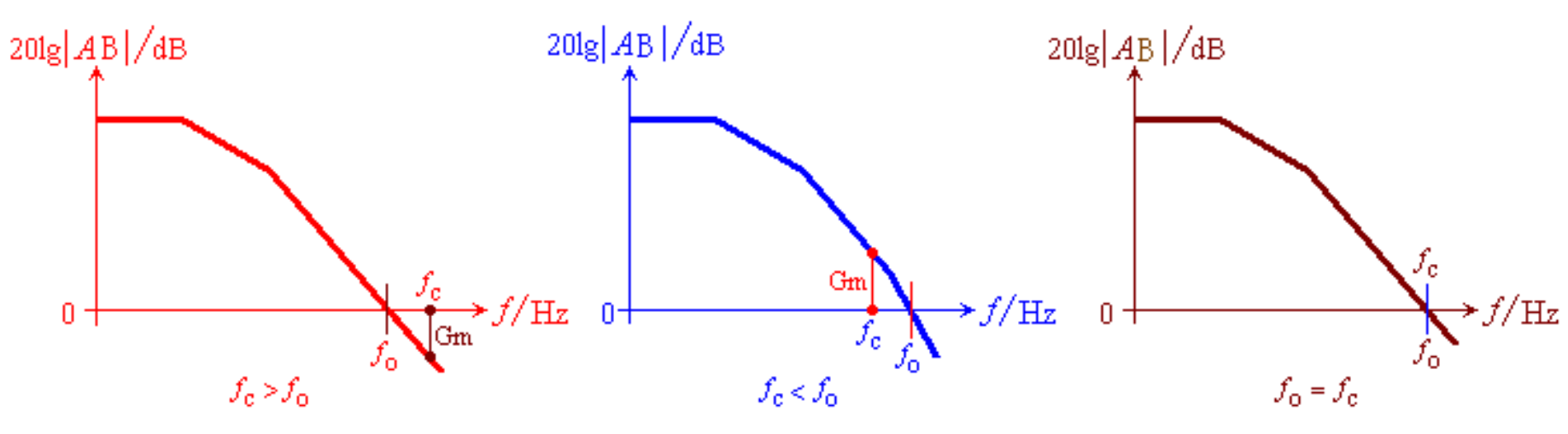
$$G_m = 20\lg|AB|_{f=f_C}$$

phase margin:

$$\phi_m = 180^0 - |\phi(f_0)|$$

Determination of Closed-Loop Stability

Determination Based on Loop Gain



(a) $f_c > f_0$, $G_m < 0$

$AB < 1$

magnitude condition isn't satisfied

stable

(b) $f_c < f_0$, $G_m > 0$

$AB > 1$

magnitude condition is satisfied

Oscillation

(c) $f_c = f_0$, $G_m = 0$

$AB = 1$

critical

Determination of Closed-Loop Stability

- **Example**

$$A_U(f) = \frac{10^4}{(1+\mathrm{j}\frac{f}{10^4})(1+\mathrm{j}\frac{f}{10^5})(1+\mathrm{j}\frac{f}{10^6})}$$

Determine the closed-loop stability?

Solution：

1. Draw the magnitude and phase frequency characteristic curves

IF voltage gain $A_U = 10^4 = 80$ dB

3 high frequency poles: $f_{p1} = 10^4$, $f_{p2} = 10^5$, $f_{p3} = 10^6$

Determination of Closed-Loop Stability

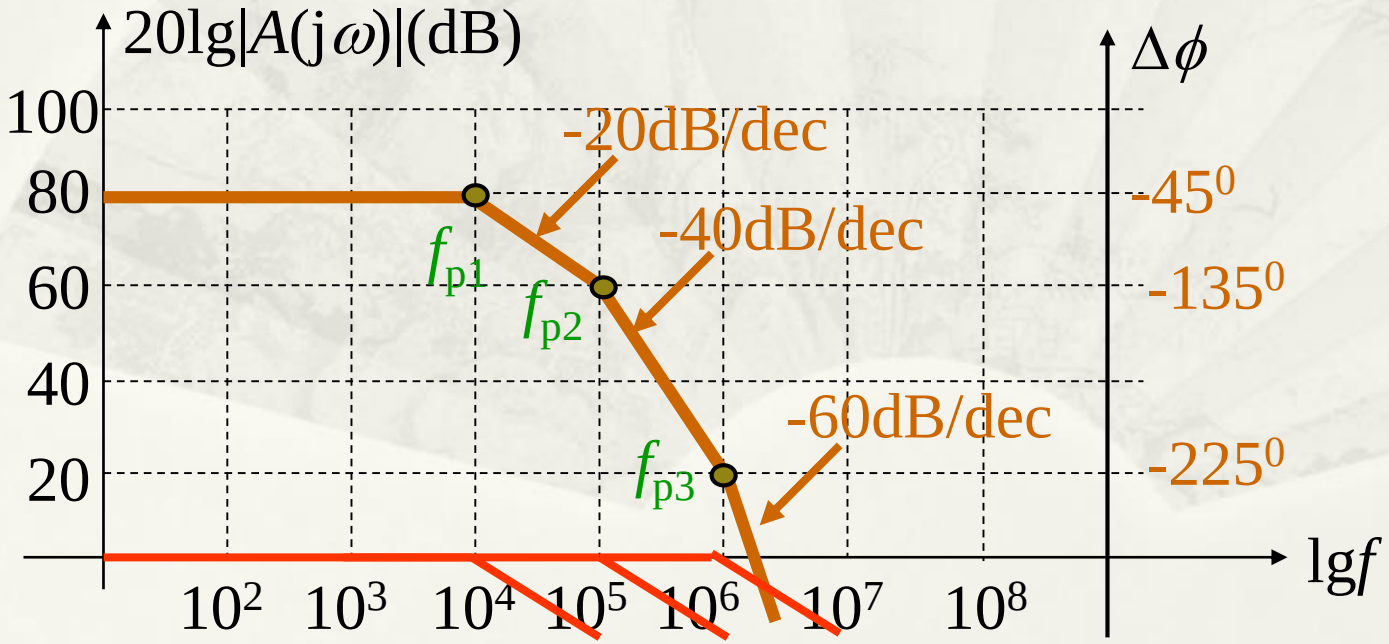
- **Example**

Solution:

1. Draw the magnitude and phase frequency characteristic curves

IF voltage gain $A_U = 10^4 = 80$ dB

3 high frequency poles: $f_{p1} = 10^4$, $f_{p2} = 10^5$, $f_{p3} = 10^6$



Determination of Closed-Loop Stability

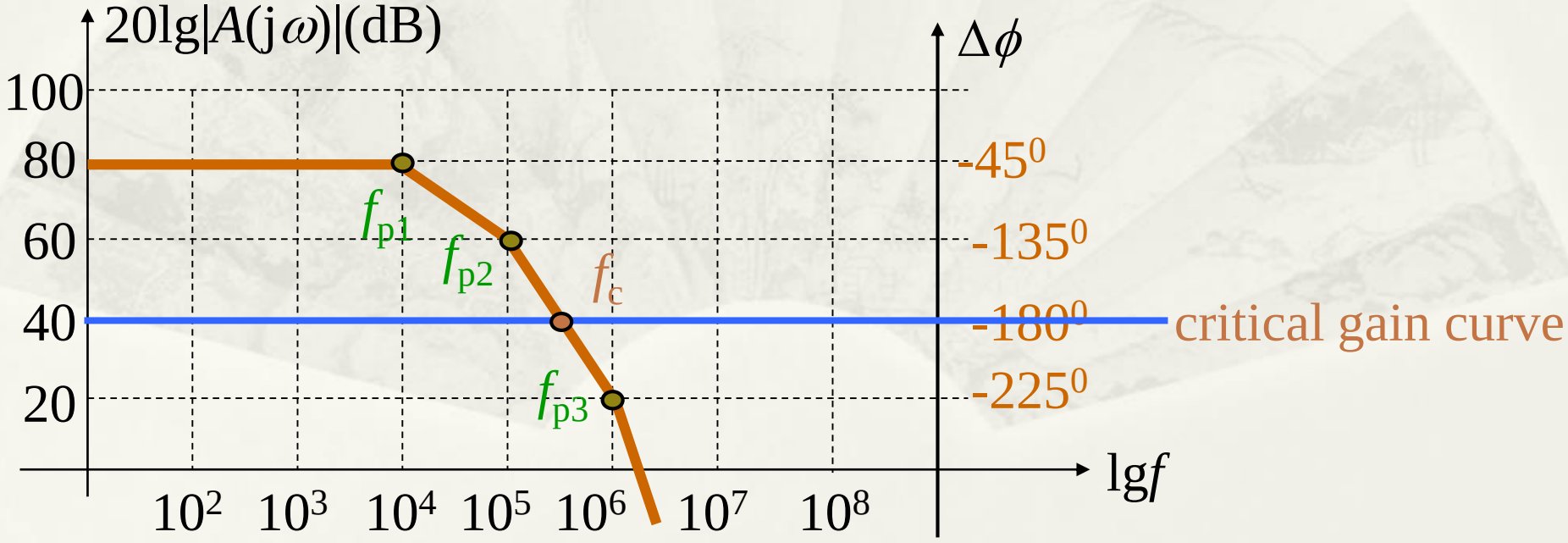
- **Example**

Solution:

2. Draw the critical gain line

✓The closed-loop gain is reduced after applying negative feedback;

✓Self-oscillation will be generated when A_{uf} is reduced to blow the critical line



Determination of Closed-Loop Stability

Example

Solution:

3. Check the magnitude condition, determine the maximum feedback coefficient.

In deep feedback, $A_{Uf} = 1/B$

$$20\lg|AB| = 20\lg|A| - 20\lg|1/B| = 40 - 20\lg|1/B| = 0\text{dB}$$

$$\Downarrow$$

$$20\lg|1/B| = 40\ \text{dB}$$

$$\Downarrow$$

$$B = 1/100 = 0.01$$

So, the critical feedback coefficient is 0.01

Determination of Closed-Loop Stability

Example

Solution：

3. Check the magnitude condition, determine the maximum feedback coefficient.

✓ The system is stable if the closed-loop gain is reduced to 60 dB after applying negative feedback.

magnitude margin: G_m = 20dB

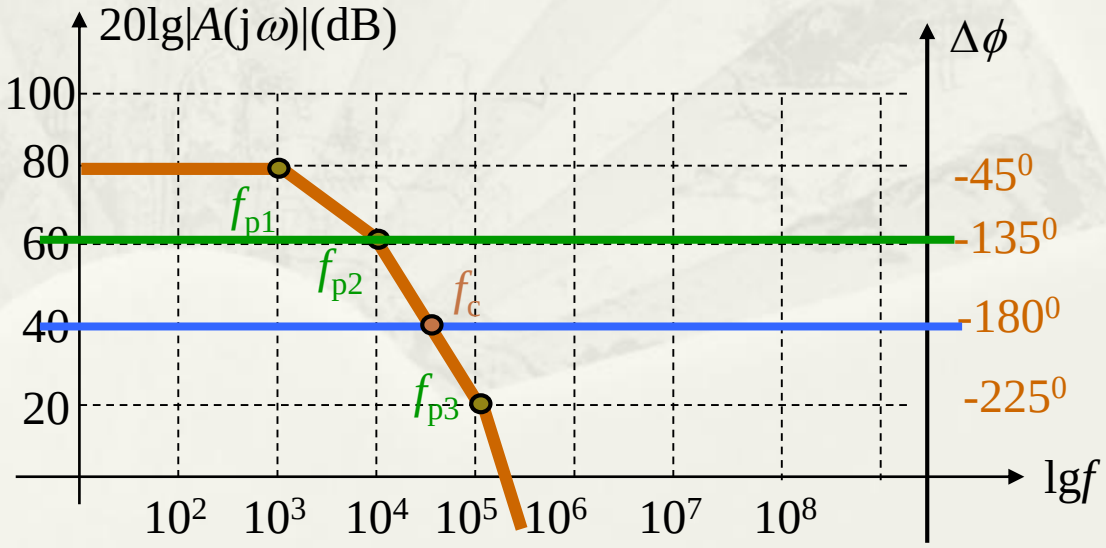
phase margin: $\phi_m = 45^0$

✓ In engineering design，margin should be left.

Generally:

$G_m > 10$dB

$\phi_m \geq 45^0$



Determination of Closed-Loop Stability

Example

Solution：

3. Check the magnitude condition, determine the maximum feedback coefficient.

✓ The system is stable if the closed-loop gain is reduced to 60 dB after applying negative feedback.

magnitude margin: G_m = 20dB

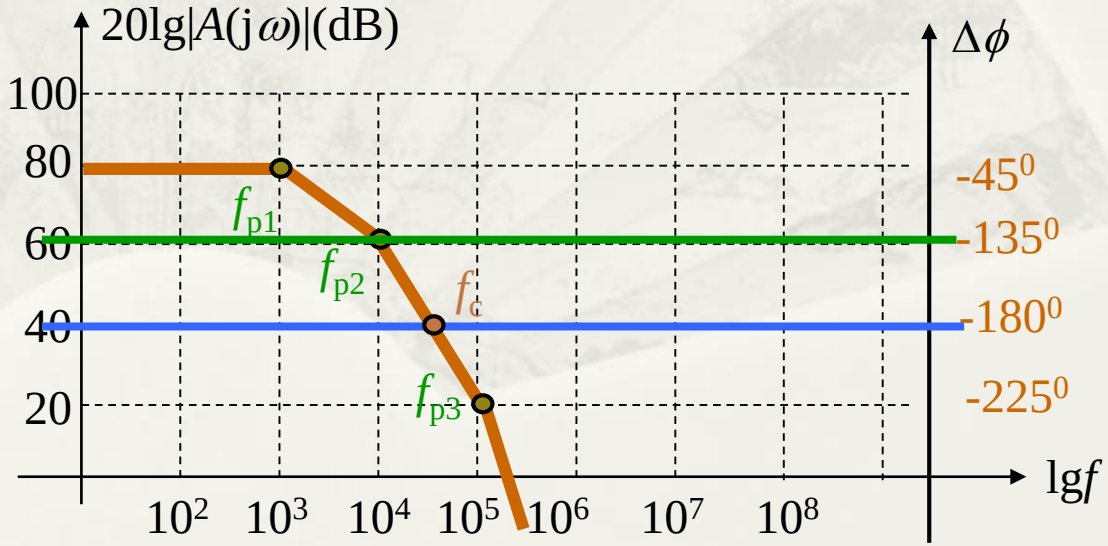
phase margin: $\phi_m = 45^0$

✓ In engineering design，margin should be left.

Generally:

$G_m > 10dB$

$\phi_m \geq 45^0$



5-5 Negative Feedback in Amplifiers

5-5-1 Basic Concepts

5-5-2 Category

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5-5-6 Brief Summary

5-5-6 Brief Summary

Category

Negative feedbacks can be divided into voltage series, voltage parallel, current series and current parallel negative feedback.

The determination method:

(1) find feedback network -- related to input and output loops.

(2) determine the type of feedback

① voltage or current feedback

It's voltage feedback if the feedback signal is zero when the output is shorted. Otherwise, it's current feedback.

② series or parallel feedback

It's parallel feedback if the feedback and input signals are added to the same node in input loop. Otherwise, it's series feedback.

5-5-6 Brief Summary

Category

The determination method:

(1) find feedback network -- related to input and output loops.

(2) determine the type of feedback

① voltage or current feedback

② series or parallel feedback

③ positive or negative feedback

Instantaneous polarity method:

- ✓ Assume the polarity of the input voltage
- ✓ Determine the instantaneous polarity of relevant nodes
- ✓ Finally, determine the instantaneous polarity of the feedback signal

It's negative feedback if the polarity of the feedback signal let the net input reduced. Otherwise, it's positive feedback.

5-5-6 Brief Summary

Category

The determination method:

(1) find feedback network -- related to input and output loops.

(2) determine the type of feedback

① voltage or current feedback

② series or parallel feedback

③ positive or negative feedback

④ DC or AC feedback

Capacitor observation method:

- ✓ It's AC feedback if there is coupling capacitors in feedback path
- ✓ It's DC feedback if there is bypass capacitors in feedback path
- ✓ It's DC-AC mixed feedback if there is no capacitors in feedback path

5-5-6 Brief Summary

Description

① Any type of negative feedback can be described by a unified feedback equation

$$A_f = \frac{A}{1 + AB}$$

② Different type can effect on the subscripts and dimensions of A, A_f and B

③ Feedback depth can effect on the performance of the feedback AMP

5-5-6 Brief Summary

- **Negative feedback is an important means to improve the performance of AMPs.**

① increase the stability of gain
- ✓ voltage negative feedback can stabilize output voltage
- ✓ current negative feedback can stabilize output voltage

② improve non-linear distortion, reduce internal noise and interference

③ change input and output resistances
- ✓ series negative feedback can increase input resistance
- ✓ parallel negative feedback can reduce input resistance
- ✓ voltage negative feedback can reduce output resistance
- ✓ current negative feedback can increase output resistance

④ expand pass-band width of AMPs

5-5-6 Brief Summary

- **Analysis Method Based on Block Diagram**

Determine the type of feedback

Determine the output
voltage feedback: $X_o=U_o$
current feedback: $X_o=I_o$

Determine the input & feedback
series feedback: X_i, X_f are voltage
parallel feedback: X_i, X_f are current

↓

Draw the basic AMP
Consider the load effects of the feedback network

↓

Calculate the parameters of the basic AMP -- A, r_i, r_o

↓

Calculate the parameters of the feedback AMP -- B, F, A_f , r_{if}, r_{of}, r'_{if}, r'_{of}

↓

calculate other parameters

5-5-6 Brief Summary

- **Self-Oscillation in multi-poles closed-loop system**

Be determined based on the bode plot of closed-loop or open-loop gains

Conditions of self-oscillation:

magnitude: $|AB|=1$

phase: $\arg A(j\omega)B(j\omega)=\pm 180^0$

HOMEWORK